

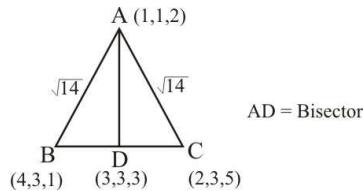
Exercise J01

Q1. The vertices of a triangle are A (1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the internal bisector of the angle A is :

- (A) $\hat{i} + \hat{j} + 2\hat{k}$ (B) $2\hat{i} - 2\hat{j} + \hat{k}$ (C) $2\hat{i} + 2\hat{j} - \hat{k}$ (D) $2\hat{i} + 2\hat{j} + \hat{k}$

Ans. D

Sol.



$$\overrightarrow{AD} = (2, 2, 1)$$

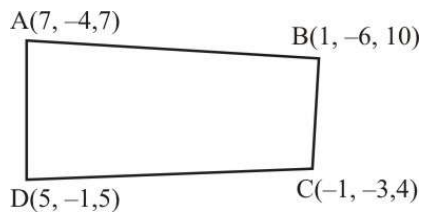
Q2. Consider points A, B, C and D with position vectors

$7\hat{i} - 4\hat{j} + 7\hat{k}$, $\hat{i} - 6\hat{j} + 10\hat{k}$, $-\hat{i} - 3\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 5\hat{k}$ respectively. Then ABCD is a-

- (A) parallelogram but not a rhombus (B) square (C) rhombus
(D) None of these

Ans. D

Sol.



$$\overrightarrow{AB} = (-6, -2, 3)$$

$$\overrightarrow{BC} = (-2, 3, -6)$$

$$\overrightarrow{CD} = (6, 2, 1)$$

$$\overrightarrow{DA} = (2, -3, 2)$$

$$|\overrightarrow{AB}| = 7, |\overrightarrow{BC}| = 7, |\overrightarrow{CD}| = \sqrt{41}, |\overrightarrow{DA}| = \sqrt{17}$$

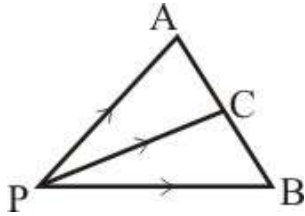
Q3. If C is the mid point of AB and P is any point outside AB, then -

- (A) $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$ (B) $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC}$ (C) $\overrightarrow{PA} + \overrightarrow{PB} + 2\overrightarrow{PC} = \overrightarrow{0}$

- (D) $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = \overrightarrow{0}$

Ans. A

Sol.



$$\begin{aligned}
 \overrightarrow{PA} + \overrightarrow{PB} &= (\overrightarrow{PA} + \overrightarrow{AC}) + (\overrightarrow{PB} + \overrightarrow{BC}) - (\overrightarrow{AC} + \overrightarrow{BC}) \\
 &= \overrightarrow{PC} + \overrightarrow{PC} - (\overrightarrow{AC} - \overrightarrow{CB}) \\
 &= 2\overrightarrow{PC} - 0 \\
 (\because \overrightarrow{AC} &= \overrightarrow{CB}) \\
 \therefore \overrightarrow{PA} + \overrightarrow{PB} &= 2\overrightarrow{PC}
 \end{aligned}$$

- Q4. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a} + 3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 2\vec{c}$ is collinear with $\vec{a}$ then $\vec{a} + 3\vec{b} + 6\vec{c}$ is
- (A) $\vec{a} + \vec{c}$ (B) $\vec{a}$ (C) $\vec{c}$ (D) $\vec{0}$

Ans. D

Sol. Let

$$\vec{a} + 3\vec{b} = \lambda\vec{c}$$

add $6\vec{c}$ both side

$$\vec{a} + 3\vec{b} + 6\vec{c} = (\lambda + 6)\vec{c}$$

Let

$$\vec{b} + 2\vec{c} = \vec{a}$$

$$3\vec{b} + 6\vec{c} = 3\vec{a}$$

$$3\vec{b} + 6\vec{c} + 3\vec{a} = 3\vec{a}$$

add $\vec{a}$ both side

$$\vec{a} + 3\vec{b} + 6\vec{c} = (3 + 1)\vec{a}$$

$$\text{Hence } \lambda + 6 = 3 + 1 = 0$$

$$\text{so } \vec{a} + 3\vec{b} + 6\vec{c} = \vec{0}$$

- Q5. If $|\vec{a}| = 5, |\vec{a} - \vec{b}| = 8$ and $|\vec{a} + \vec{b}| = 10$, then $|\vec{b}|$ is equal to :

- (A) 1 (B) $\sqrt{57}$ (C) 3 (D) None of these

Ans. B

Sol. $\vec{a} = 5, |\vec{a} + \vec{b}| = 10$

$$a^2 + b^2 - 2\vec{a} \cdot \vec{b} = 64 \dots\dots\dots(1)$$

$$a^2 + b^2 - 2 + 2\vec{a} \cdot \vec{b} = 100 \dots\dots\dots(2)$$

$$(1) + (2)$$

$$2(a^2 + b^2) = 164$$

$$b^2 = 82 - a^2$$

$$b^2 = 82 - 25$$

$$|a| = \sqrt{57}$$

Q6. Angle between diagonals of a parallelogram whose side are represented by

$$\vec{a} = 2\hat{i} + \hat{j} + \hat{k} \text{ and } \vec{b} = 2\hat{i} - \hat{j} - \hat{k}$$

- (A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(\frac{4}{9}\right)$ (D) $\cos^{-1}\left(\frac{5}{9}\right)$

Ans. A

Sol.

Diagonals of a parallelogram are

$$\vec{\lambda}_1 = \vec{a} - \vec{b} = (3, 0, 0)$$

$$\vec{\lambda}_2 = \vec{a} + \vec{b} = (1, 2, 2)$$

$$\cos \theta = \frac{\vec{\lambda}_1 \cdot \vec{\lambda}_2}{|\vec{\lambda}_1| |\vec{\lambda}_2|} = \frac{3}{3 \cdot \sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$= \frac{(10 - 3 + 2)}{(35)} \sqrt{5}$$

$$= \frac{9}{35} (5\hat{i} - 3\hat{j} + \hat{k})$$

Q7. Two adjacent sides of a parallelogram ABCD are given by

$$\vec{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k} \text{ and } \vec{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB, then the cosine of the angle α is given by

- (A) $\frac{8}{9}$ (B) $\frac{\sqrt{17}}{9}$ (C) $\frac{1}{9}$ (D) $\frac{4\sqrt{5}}{9}$

Ans. B

Sol. T Let θ be the angle between $\overrightarrow{AB}$ and $\overrightarrow{AD}$

$$\Rightarrow \theta + \alpha = 90$$

$$\Rightarrow \alpha = 90 - \theta$$

$$\Rightarrow \cos \alpha = \sin \theta \quad \dots\dots\dots(i)$$

$$\text{Now, } \cos \theta = \frac{\overrightarrow{AB} \cdot \overrightarrow{AD}}{|\overrightarrow{AB}| |\overrightarrow{AD}|} = \frac{8}{9}$$

$$\Rightarrow \cos \theta = \frac{\sqrt{17}}{9} \text{ from (i)}$$

Q8. Let two non-collinear unit vector $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O, let M be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$. Then,

- (A) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$ (B) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$
 (C) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$ (D) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

Ans. A

Sol. $|\overrightarrow{OP}| = |\hat{a} \cos t + \hat{b} \sin t|$
 $= (\cos^2 t + \sin^2 t + 2 \sin t \cos t \hat{a} \cdot \hat{b})^{1/2}$
 $= (1 + \sin 2t \hat{a} \cdot \hat{b})^{1/2}$
 $\therefore |\overrightarrow{OP}|_{\max} = (1 + \hat{a} \cdot \hat{b})^{1/2}, \text{ when } t = \frac{\pi}{4}$

$$\text{Now } \hat{u} = \frac{\hat{a} + \hat{b}}{\sqrt{2} \frac{|\hat{a} + \hat{b}|}{\sqrt{2}}} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$$

Q9. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $\vec{c} \cdot \vec{d} = 0$

$$\Rightarrow (\hat{a} + 2\hat{b}) \cdot (5\hat{a} - 4\hat{b}) = 0$$

$$\Rightarrow 5 - 8 + 6\hat{a} \cdot \hat{b} = 0$$

$$\Rightarrow \hat{a} \cdot \hat{b} = 1/2$$

$$\Rightarrow \cos\theta = 1/2$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

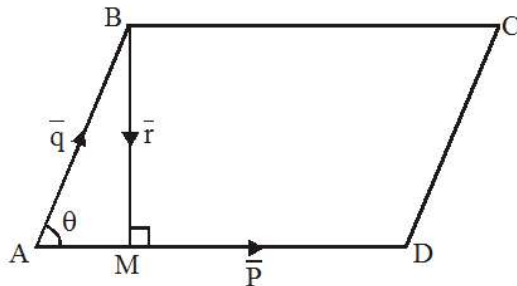
Q10. Let ABCD be a parallelogram such that $\vec{AB} = \vec{q}$, $\vec{AD} = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by :

(A) $\vec{r} = -3\vec{q} + \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$ (B) $\vec{r} = 3\vec{q} - \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$ (C) $\vec{r} = -\vec{q} + \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}\right)\vec{p}$

(D) $\vec{r} = \vec{q} - \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}\right)\vec{p}$

Ans. C

Sol.



$$\vec{q} + \vec{r} = \vec{AM}$$

$$\Rightarrow \vec{r} = -\vec{q} + \vec{AM}$$

$$\Rightarrow \vec{r} = -\vec{q} + \frac{\vec{p} \cdot \vec{q}}{|\vec{p}|} \vec{p}$$

$$\Rightarrow \vec{r} = -\vec{q} + \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}\right)\vec{p}$$

Q11. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u} \cdot \vec{b} = 24$, then $|\vec{u}|^2$ is equal to

(A) 256 (B) 84 (C) 336 (D) 315

Ans. C

Sol. $\vec{u} = \lambda (\vec{a} \times (\vec{a} \times \vec{b}))$

$$= \lambda ((\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b})$$

$$= \lambda (2\vec{a} - 14\vec{b})$$

$$\vec{u} = \lambda (4\hat{i} - 8\hat{j} - 16\hat{k})$$

$$\vec{u} \cdot \vec{b} = 24 \Rightarrow \lambda = -1$$

$$\Rightarrow \vec{u} = -4\hat{i} + 8\hat{j} + 16\hat{k}$$

$$\Rightarrow |\vec{u}|^2 = 16 + 64 + 256 = 336$$

Q12. Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is :

- (A) $-\hat{i} + \hat{j} + 2\hat{k}$ (B) $3\hat{i} - \hat{j} + \hat{k}$ (C) $3\hat{i} + \hat{j} - \hat{k}$ (D) $\hat{i} - \hat{j} - \hat{k}$

Ans. C

Sol. $\vec{a} = \hat{i} + \hat{j}$ & $\vec{b} = 2\hat{i} - \hat{k}$

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \Rightarrow (\vec{r} - \vec{b}) \times \vec{a} = 0$$

$$\Rightarrow \vec{r} = \vec{b} + \lambda \vec{a} \dots\dots\dots(i)$$

similarly $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$$\Rightarrow \vec{r} = \vec{a} + \mu \vec{b} \dots\dots\dots(ii)$$

Putting the vector $\vec{a}$ & $\vec{b}$ in (i) & (ii) and equating we get

$$2\hat{i} - \hat{k} + \lambda(\hat{i} + \hat{j}) = \hat{i} + \hat{j} + (2\hat{i} - \hat{k})$$

$$\Rightarrow 2 + \lambda = 1 + 2, \lambda = 1, \mu = 1$$

$\therefore$ Point of intersection is $3\hat{i} + \hat{j} - \hat{k}$.

Q13. Unit vector perpendicular to the plane of the triangle ABC with position vectors $\vec{a}, \vec{b}, \vec{c}$ of the vertices A, B, C is:

- (A) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{\Delta}$ (B) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{2\Delta}$ (C) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{4\Delta}$

(D) None of these

Ans. B

Sol.

$$\vec{n} = \frac{\vec{AB} \times \vec{AC}}{AB \cdot AC} = \frac{(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})}{2\Delta}$$

$$= \frac{\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{c} \times \vec{a}}{2\Delta}$$

$$= \frac{\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}}{2\Delta}$$

Q14. If $\vec{u}$ and $\vec{v}$ are unit vectors and θ is the acute angle between them, then $2\vec{u} \times 3\vec{v}$ is a unit vector for –

- (A) Exactly two values of θ (B) More than two values of θ (C) No value of θ
 (D) Exactly one value of θ

Ans. D

Sol.

$$|\vec{\mu}| = |\vec{\mu}| = 1$$

$$|2\vec{\mu} \times 3\vec{v}| \sin \theta = 1$$

$$\sin \theta = \frac{1}{6}$$

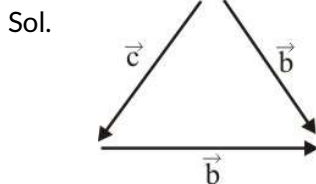
$\therefore \theta$ is acute so

exactly one value of θ

Q15. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct?

- (A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$ (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$ (C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$
 (D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

Ans. B



$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

with $\vec{a}$

$$\vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{0}$$

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{b}$$

$$\text{Hence } \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$$

bc $\vec{a}, \vec{b}, \vec{c}$ are unit vector and Non-parallel vector.

Q16. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then $(\lambda, \mu) =$

- (A) $(-3, 2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(3, -2)$

Ans. A

Sol. Give $\vec{a} \perp \vec{b}$, $\vec{a} \perp \vec{c}$ & $\vec{b} \perp \vec{c}$

$$\text{so } \vec{a} \cdot \vec{c} = 0 \text{ \& } \vec{b} \cdot \vec{c} =$$

$$\Rightarrow \lambda - 1 + 2 = 0 \text{ \& } 2\lambda + 4 + \mu = 0$$

$$\Rightarrow \lambda = -3 \text{ \& } \mu = 2$$

Q17. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying :
 $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 0$. Then the vector $\vec{d}$ is equal to :

- (A) $\vec{b} + \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{c}$ (B) $\vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$ (C) $\vec{b} - \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{c}$ (D) $\vec{c} + \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$

Ans. B

Sol. $\vec{a} \cdot \vec{b} \neq 0$

$$\vec{a} \cdot \vec{d} = 0$$

$$\vec{b} \cdot \vec{c} = \vec{b} \cdot \vec{d}$$

$$\vec{a} \cdot (\vec{b} \cdot \vec{c}) = \vec{a} \cdot (\vec{b} \cdot \vec{d})$$

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{d} \quad \{\vec{a} \cdot \vec{d} = 0\}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{d} = (\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b} \quad (\text{divided by } \vec{a} \cdot \vec{b})$$

$$\vec{d} = \vec{c} - \frac{(\vec{a} \cdot \vec{c})}{(\vec{a} \cdot \vec{b})} \vec{b}$$

Q18. The value of $[(\vec{a} + 2\vec{b} - \vec{c}) \cdot (\vec{a} - \vec{b}), (\vec{a} - \vec{b} - \vec{c})]$ is equal to the box product:

- (A) $[\vec{a}\vec{b}\vec{c}]$ (B) $2[\vec{a}\vec{b}\vec{c}]$ (C) $3[\vec{a}\vec{b}\vec{c}]$ (D) $4[\vec{a}\vec{b}\vec{c}]$

Ans. C

Sol. $[(\vec{a} + 2\vec{b} \times \vec{c}) \times (\vec{a} - \vec{b})] \cdot (\vec{a} - \vec{b} - \vec{c})$

$$[0 + 2\vec{a} \times \vec{a} + \vec{a} \times \vec{c} - \vec{a} \times \vec{b} + \vec{c} \times \vec{b} \cdot (\vec{a} - \vec{b} - \vec{c})]$$

$$[-3\vec{a} \times \vec{b} - \vec{b} \times \vec{c} - \vec{c} \times \vec{a}] \cdot (\vec{a} - \vec{b} - \vec{c})$$

$$[-[\vec{b}\vec{c}\vec{a}] + [\vec{c}\vec{a}\vec{b}] + 3[\vec{a}\vec{b}\vec{c}]]$$

$$= [\vec{abc}]$$

Q19.

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$ then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ is equal to :

- (A) 2 (B) 4 (C) 16 (D) 64

Ans. C

Sol.
$$\begin{bmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{bmatrix} = [\vec{abc}]^2$$

$$[\vec{abc}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= 1(1 - 2) - 1(-1 - 1) + 1(2 + 1)$$

$$= -1 + 2 + 3$$

$$= 4$$

$$[\vec{abc}]^2 = 16$$

Q20. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors

$\vec{a} + 2\vec{b} + 3\vec{c}, \lambda\vec{b} + 4\vec{c}$ and $(2\lambda - 1)\vec{c}$ are non-coplanar for

- (A) all values of λ (B) all except one value of λ (C) all except two values of λ
 (D) no value of λ

Ans. C

Sol. Let $\vec{\alpha} = \vec{a} + 2\vec{b} + 3\vec{c}$, $\vec{\beta} = \lambda\vec{b} + 4\vec{c}$ and $\vec{\gamma} = (2\lambda - 1)\vec{c}$

Then, $[\vec{\alpha}\vec{\beta}\vec{\gamma}] = \begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & (2\lambda - 1) \end{vmatrix} [\vec{abc}]$

$$\Rightarrow [\vec{abc}] = \lambda(2\lambda - 1)[\vec{abc}]$$

$$\Rightarrow [\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0, \text{ if } \lambda = 0, \frac{1}{2} [\because [\vec{abc}] \neq 0]$$

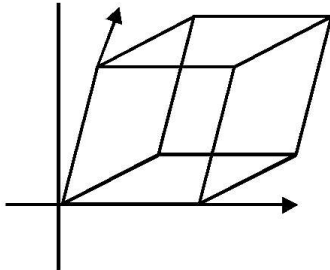
Hence, $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ are non-coplanar for all values of λ except two values 0 and $\frac{1}{2}$.

Q21. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$. Then the volume of the parallelopiped is

- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{3}}$

Ans. A

Sol.



Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{a} = \hat{i} \text{ then } \vec{b} = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$$

$$\text{so } x = \frac{1}{2}$$

$$\& \frac{x}{2} + \frac{y\sqrt{3}}{2} = \frac{1}{2}$$

$$\Rightarrow y\sqrt{3} = \frac{1}{2} \quad \therefore y = \frac{1}{2\sqrt{3}}$$

$$\text{also } z^2 = 2/3 \Rightarrow z = \sqrt{2/3}$$

$$\text{so volume} = \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & \pm\sqrt{\frac{2}{3}} \end{vmatrix} = \frac{1}{\sqrt{2}}$$

Alternative

$$\text{volume } |\vec{a} \cdot (\vec{b} \times \vec{c})| = \sqrt{\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}} = \sqrt{\begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}} = \frac{1}{\sqrt{2}}$$

Q22. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ & $\vec{c} = \hat{j} + \hat{k}$ & bisects the angle between $\vec{b}$ & $\vec{c}$. Then which one of the following gives possible values of α & β ?

- (A) $\alpha = 1, \beta = 2$ (B) $\alpha = 2, \beta = 1$ (C) $\alpha = 1, \beta = 1$ (D) $\alpha = 2, \beta = 2$

Ans. C

Sol. $\vec{a} = \lambda(\hat{b} + \hat{c})$

$$\Rightarrow \alpha\hat{i} + 2\hat{j} + \beta\hat{k} = \lambda \left(\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{2}} \right)$$

$$\lambda = \sqrt{2}\alpha \text{ and } \lambda = \sqrt{2} \text{ and } \lambda = \sqrt{2}\beta$$

$$\Rightarrow \alpha = 1 \text{ and } \beta = 1$$

Q23. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $[3\vec{u} \vec{p}\vec{v} \vec{p} \vec{w}] - [p\vec{v} \vec{w} \vec{q}\vec{u}] - [2\vec{w} \vec{q}\vec{v} \vec{q}\vec{u}] = 0$ holds for :

(A) exactly two values of (p,q) (B) more than two but not all values of (p, q)

(C) all values of (p, q) (D) exactly one value of (p, q)

Ans. D

Sol. $[3\vec{u} \vec{p}\vec{v} \vec{p}\vec{w}] - [p\vec{v} \vec{w} \vec{q}\vec{u}] - [2\vec{w} \vec{q}\vec{v} \vec{q}\vec{u}] = 0$

$$3p^2 [\vec{u}\vec{v}\vec{w}] - pq[\vec{v}\vec{w}\vec{u}] - 2q^2 [\vec{w}\vec{v}\vec{u}] = 0$$

$$(3p^2 - pq + 2q^2) \cdot [\vec{u}\vec{v}\vec{w}] = 0$$

$$3p^2 - pq + 2q^2 = 0$$

has exactly one solution

$$p = q = 0$$

Q24. If the vectors $p\hat{i} + \hat{j} + \hat{k}, \hat{i} + q\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + r\hat{k}$ ($p \neq q \neq r \neq 1$) are coplanar, then the value of $pqr - (p+q+r)$ is

$pqr - (p + q + r)$ is

(A) -2 (B) 2 (C) 0 (D) -1

Ans. A

Sol. $\begin{vmatrix} p & 1 & 1 \\ 1 & q & 1 \\ 1 & 1 & r \end{vmatrix} = 0$

$$p(qr - 1) - (r - 1) + (1 - q) = 0$$

$$pqr - p - r + 1 + 1 - q = 0$$

$$pqr - (p + r + q) + 2 = 0$$

$$pqr - (p + r + q) = -2$$

Q25. If $\vec{b}$ and $\vec{c}$ are two non-collinear vectors such that $\vec{a} \parallel (\vec{b} \times \vec{c})$, then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ is equal to

(A) $\vec{a}^2(\vec{b} \cdot \vec{c})$ (B) $\vec{b}^2(\vec{a} \cdot \vec{c})$ (C) $\vec{c}^2(\vec{a} \cdot \vec{b})$ (D) None of these

Ans. A

Sol. $\vec{a} \parallel (\vec{b} \times \vec{c})$

$$\vec{a} \perp \vec{b}, \vec{a} \perp \vec{c}$$

$$\text{Let } \vec{a} \times \vec{b} = \vec{P}$$

$$\vec{P} \cdot (\vec{a} \times \vec{c})$$

$$= (\vec{P} \times \vec{a}) \cdot \vec{c}$$

$$= ((\vec{a} \times \vec{b}) \times \vec{a}) \cdot \vec{c}$$

$$= [\vec{b}(\vec{a} \cdot \vec{a}) - \vec{a}(\vec{a} \cdot \vec{b})] \cdot \vec{c}$$

$$= (\vec{b} \cdot \vec{c}) (\vec{a})^2$$

Q26. Vector of length 3 unit which is perpendicular to $\hat{i} + \hat{j} + \hat{k}$ and lies in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + 3\hat{j}$

$$(A) \frac{3}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k}) \quad (B) \frac{3}{\sqrt{6}}(2\hat{i} - \hat{j} - \hat{k}) \quad (C) \frac{3}{\sqrt{114}}(8\hat{i} - 7\hat{j} - \hat{k}) \quad (D) \frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$$

Ans. D

Sol. Let vector is $\vec{P}$

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}, \quad \vec{b} = \hat{i} + \hat{j} + \hat{k}, \quad \vec{c} = 2\hat{i} - 3\hat{j}$$

$$\vec{P} \perp \vec{a}, \quad \vec{P} \perp (\vec{b} \times \vec{c})$$

$$\vec{P} = \lambda(\vec{a} \times (\vec{b} \times \vec{c}))$$

$$\vec{P} = \lambda(\vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}))$$

$$\vec{P} = \lambda(\vec{b}(-1) - \vec{c}(3))$$

$$\vec{P} = \lambda(-\hat{i} - \hat{j} - \hat{k} - 6\hat{i} + 9\hat{j})$$

$$\vec{P} = \lambda(-7\hat{i} + 8\hat{j} - \hat{k})$$

$$|\vec{P}| = 3 = \lambda \sqrt{49 + 64 + 1}$$

$$\lambda = \frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$$

Q27. If $[\vec{a}\vec{b}\vec{c}] = 4$ then $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] =$

(A) 4 (B) 2 (C) 8 (D) 16

Ans. D

$$\begin{aligned}\text{Sol. } [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] &= [\vec{abc}]^2 \\ &= (4)^2 \\ &= 16\end{aligned}$$

Q28. If $\vec{c} = 2\lambda(\vec{a} \times \vec{b}) + 3\mu(\vec{b} \times \vec{a})$; $\vec{a} \times \vec{b} \neq 0$, $\vec{c} \cdot (\vec{a} \times \vec{b}) = 0$ then-

(A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) None of these

Ans. B

$$\begin{aligned}\text{Sol. } \vec{C} &= 2\lambda(\vec{a} \times \vec{b}) + 3\mu(\vec{b} \times \vec{a}) \quad \vec{a} \times \vec{b} \neq 0; \\ \vec{C} &= (\vec{a} \times \vec{b}) = 0 \\ (\vec{c} \times \vec{a}) \cdot \vec{b} &= 0 \\ [2\lambda(\vec{a} \times \vec{b}) \times \vec{a} + 3\mu(\vec{b} \times \vec{a}) \times \vec{a}] \cdot \vec{b} &= 0 \\ 2\lambda(\vec{b} \cdot \vec{b})(\vec{a})^2 - 2\lambda(\vec{a} \cdot \vec{b})^2 + 3\mu(\vec{a} \cdot \vec{b})^2 - 3\mu(\vec{a})(\vec{b}) &= 0 \\ (2\lambda - 3\mu)\vec{a}^2\vec{b}^2 + 3(3\mu - \lambda)(\vec{a} \cdot \vec{b})^2 &= 0 \\ (2\lambda - 3\mu)((\vec{a})^2(\vec{b})^2 - (\vec{a} \cdot \vec{b})^2) &= 0 \\ (2\lambda - 3\mu)(|\vec{a} \times \vec{b}|^2) &= 0 \\ 2\lambda - 3\mu &= 0\end{aligned}$$

Q29. For any vector $\vec{a}$, $|\vec{a} \times \vec{i}|^2 + |\vec{a} \times \vec{j}|^2 + |\vec{a} \times \vec{k}|^2$ is equal to –

(A) $|\vec{a}|^2$ (B) $2|\vec{a}|^2$ (C) $3|\vec{a}|^2$ (D) None of these

Ans. B

$$\begin{aligned}\text{Sol. Let } \vec{a} &= a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \\ (\vec{a} \times \hat{i}) &= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times \hat{i} \\ &= -\hat{k}a_2 + a_3\hat{j} \\ &= a_3\hat{j} - a_2\hat{k} \\ |\vec{a} \times \hat{i}|^2 &= (\sqrt{a_1^2 + a_2^2})^2 = a_2^2 + a_3^2 \dots\dots\dots(1)\end{aligned}$$

Similarly

$$|\vec{a} \times \hat{j}|^2 = (\sqrt{a_1^2 + a_2^2})^2 = a_1^2 + a_2^2 \dots\dots\dots(2)$$

$$|\vec{a} \times \hat{k}|^2 = a_1^2 + a_2^2 \dots\dots\dots(3)$$

$$(1) + (2) + (3)$$

$$= 2(a_1^2 + a_2^2 + a_3^2)$$

$$= 2|\vec{a}|^2$$

Q30. If $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$, where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a} \cdot \vec{b} \neq 0, \vec{b} \cdot \vec{c} \neq 0$, then $\vec{a}$ and $\vec{c}$ are –

- (A) inclined at an angle of $\pi/6$ between them (B) perpendicular (C) parallel
(D) inclined at an angle of $\pi/3$ between them

Ans. C

Sol. $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$

$$\Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

$$\Rightarrow (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{b})\vec{c}$$

So that $\vec{a}$ is parallel to $\vec{c}$

Q31. If $\vec{a} = \frac{1}{\sqrt{10}}(3\vec{i} + \vec{k})$ and $\vec{b} = \frac{1}{7}(2\vec{i} + 3\vec{j} - 6\vec{k})$, then the value of $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$ is

- (A) 5 (B) 3 (C) -5 (D) -3

Ans. C

Sol. $\vec{a} \cdot \vec{b} = 0$ and $|\vec{a}| = |\vec{b}| = 1$

$$(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b}) = (\vec{a} \times \vec{b}) \times \vec{a} + (\vec{a} \times \vec{b}) \times 2\vec{b}$$

$$= -[\vec{a} \times (\vec{a} \times \vec{b}) + 2\vec{b} \times (\vec{a} \times \vec{b})] \times 2\vec{b}$$

$$= -[(\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} + 2 \times (\vec{b} \cdot \vec{b})\vec{a} - 2(\vec{b} \cdot \vec{a})\vec{b}]$$

$$= -[0 - \vec{b} + 2\vec{a} + 0] = [\vec{b} - 2\vec{a}]$$

$$= (2\vec{a} - \vec{b}) \cdot [(\vec{a} - \vec{b}) \times (\vec{a} + 2\vec{b})]$$

$$= (2\vec{a} - \vec{b}) \cdot (\vec{b} - 2\vec{a})$$

$$= 4\vec{a} \cdot \vec{a} - \vec{b} \cdot \vec{b} + 4\vec{a} \cdot \vec{b} = -5$$

- Q32. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin \theta$ is :
- (A) $\frac{2\sqrt{2}}{3}$ (B) $-\frac{\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $-\frac{2\sqrt{2}}{3}$

Ans. A

Sol. $(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$

$$-\vec{b} \cdot \vec{c} = \frac{1}{3} (\vec{b}) (\vec{c})$$

$$\cos \theta = -\frac{1}{3}$$

$$\sin \theta = \frac{2\sqrt{2}}{3}$$

- Q33. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\sqrt{3}}{2} (\vec{b} + \vec{c})$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is :

- (A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = \frac{\sqrt{3}}{2} \vec{b} + \frac{\sqrt{3}}{2} \vec{c}$

$$-\vec{a} \cdot \vec{b} = \frac{\sqrt{3}}{2}$$

$$-\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{5\pi}{6}$$

- Q34. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. Let $\vec{c}$ be a vector such that $|\vec{c} - \vec{a}| = 3$ $|(\vec{a} - \vec{b}) \times \vec{c}| = 3$, and the angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ be 30° . Then $\vec{a} \cdot \vec{c}$ is equal to

- (A) 5 (B) $\frac{1}{8}$ (C) $\frac{25}{8}$ (D) 2

Ans. D

Sol. $\vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k} \Rightarrow |\vec{a} \times \vec{b}| = 3 \quad |\vec{c} - \vec{a}| = 3$

$$|\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ = 3 \quad c^2 + a^2 - 2\vec{a} \cdot \vec{c} = 9$$

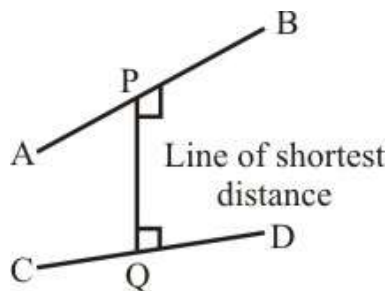
$$|\vec{c}| = 2 \quad \vec{c} \cdot \vec{a} = 2$$

- Q35. A line with direction cosines proportional to 2, 1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The coordinates of each of the points of intersection are given by-

- (A) $(3a, 3a, 3a), (a, a, a)$ (B) $(3a, 2a, 3a), (a, a, a)$ (C) $(3a, 2a, 3a), (a, a, 2a)$
 (D) $(2a, 3a, 3a), (2a, a, a)$

Ans. B

Sol.



Let the two lines be AB and CD having equation

$$\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = \lambda \text{ and } \frac{x+a}{2} = \frac{y}{1} = \frac{z}{1} =$$

then $P \equiv (\lambda, \lambda - a, \lambda)$ and $Q = (2 - a, ,)$

So according to question,

$$\frac{\lambda - 2 + a}{2} = \frac{\lambda - a -}{1} = \frac{-\lambda}{2}$$

$$\Rightarrow = a \text{ and } \lambda = 3$$

$$\therefore P \equiv (3a, 2a, 3a)$$

Q36. If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is-

- (A) $\pi/6$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/2$

Ans. D

Sol. $l = \cos \frac{\pi}{4}, m = \cos \frac{\pi}{4}$

we know that $l^2 + m^2 + n^2 = 1$

$$\frac{1}{2} + \frac{1}{2} + n^2 = 1 \Rightarrow n = 0$$

Hence angle with positive direction of z-axis is

$$\frac{\pi}{2}$$

Q37. The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. Solving two equation

$$1 = -(m + n) \text{ then } (m + n)^2 = m^2 + n^2$$

$$\Rightarrow mn = 0$$

If $m = 0$, If $n = 0$

$$1 = -1 \quad 1 = -m$$

hence dr are 1, 0, -1 and 1, -1, 0

$$\cos \theta$$

$$\theta = \frac{\pi}{3}$$

Q38. The two lines $x = ay + b$, $z = cy + d$; and $x = a'y + b'$, $z = c'y + d'$ are perpendicular to each other if –

$$(A) aa' + cc' = 1 \quad (B) \frac{a}{a'} + \frac{c}{c'} = -1 \quad (C) \frac{a}{a'} + \frac{c}{c'} = 1 \quad (D) aa' + cc' = -1$$

Ans. D

$$\text{Sol. } l_1 - am_1 = 0 \text{ and } cm_1 - n_1 = 0 \Rightarrow \frac{l_1}{a} = \frac{m_1}{1} = \frac{n_1}{c}$$

$$\text{Also } l_1 - a'm_2 = 0 \text{ and } c'm_2 - n_2 = 2$$

$$\Rightarrow \frac{l_1}{a'} = \frac{m_2}{1} = \frac{n_2}{c'}$$

$$\therefore l_1 l_2 + m_1 m_2 + n_1 n_2 = aa' + cc' + 1 = 0$$

Q39. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is equal to-

$$(A) 5 \quad (B) 2 \quad (C) -2 \quad (D) -5$$

Ans. D

$$\text{Sol. } \frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3} \text{ and } \frac{x-2}{3} = \frac{y-3}{3} = \frac{z-1}{2}$$

Since lines intersect in a point

$$\begin{vmatrix} k & 2 & 3 \\ 3 & k & 2 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

$$\therefore 2k^2 + 5k - 25 = 0$$

$$k = -5, 5/2$$

Q40. Equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and, is

$$(A) x + 2y - 2z = 0 \quad (B) 3x + 2y - 2z = 0 \quad (C) x - 2y + z = 0 \quad (D) 5x + 2y - 4z = 0$$

Ans. C

Sol. Normal vector to the plane containing the line $\frac{x}{3} = \frac{y}{4}$ and $\frac{x}{y} = \frac{y}{2} \frac{z}{3}$ is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 2 \\ 4 & 4 & 3 \end{vmatrix} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Let direction ratios of required plane be a, b, c.

$$\text{Now } 8a - b - 10c = 0$$

$$\text{and } 2a + 3b + 4c = 0$$

($\because$ plane contains the line $\frac{x}{2} = \frac{y}{3} \frac{z}{4}$)

$$\Rightarrow \frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$$

$$\Rightarrow \text{equation of plane is } x - 2y + z = d$$

$\because$ plane contains the line, which passes through origin, hence origin lies on a plane.

$$\Rightarrow \text{equation of required plane is } x - 2y + z = 0$$

Q41. A line with positive direction cosines passes through the point P(2, -1, 2) and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals

$$(A) 1 \quad (B) \sqrt{2} \quad (C) \sqrt{3} \quad (D) 2$$

Ans. C

Sol. Let DC's be $(\cos \alpha, \cos \alpha, \cos \alpha)$

$$3 \cos^2 \alpha = 1$$

$$\cos \alpha = \frac{1}{\sqrt{3}}$$

$$\text{Line PQ is } \frac{x-2}{1/\sqrt{3}} = \frac{y+1}{1/\sqrt{3}} = \frac{z-2}{1/\sqrt{3}} = \lambda$$

$$\left(\frac{\lambda}{\sqrt{3}} + 2, \frac{\lambda}{\sqrt{3}} - 1, \frac{\lambda}{\sqrt{3}} + 2 \right)$$

Putting in plane

$$\frac{2\lambda}{\sqrt{3}} + 4 + \frac{\lambda}{\sqrt{3}} - 1 + \frac{\lambda}{\sqrt{3}} + 2 = 9$$

$$\frac{4\lambda}{\sqrt{3}} = 4$$

$$\lambda = \sqrt{3}$$

$$Q = (3, 0, 3)$$

$$(PQ)^2 = \sqrt{3}$$

Q42. Two systems of rectangular axes have the same origin. If a plane makes intercepts a, b, c and a', b', c' on the two systems of axes respectively, then

$$(A) a^2 + b^2 + c^2 = a'^2 + b'^2 + c'^2 \quad (B) \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a'} + \frac{1}{b'} + \frac{1}{c'}$$

$$(C) \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2} \quad (D) \frac{1}{a^2 - a'^2} + \frac{1}{b^2 - b'^2} + \frac{1}{c^2 - c'^2} = 0$$

Ans. C

Sol. Equation plane in both system

$$\frac{x}{a} + \frac{4}{b} + \frac{7}{c} = 1 \quad \frac{x}{a'} + \frac{4}{b'} + \frac{7}{c'} = 1$$

Both plane are equidistance from origin

$$\left| \frac{-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} \right| : \left| \frac{-1}{\sqrt{\frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}}} \right|$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$$

Q43. Let $P(3, 2, 6)$ be a point in space and Q be a point on the line

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$$

Then the value of μ for which the vector $\vec{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is

$$(A) 1/4 \quad (B) -1/4 \quad (C) 1/8 \quad (D) -1/8$$

Ans. A

Sol. Let Q be $(1 - 3\mu, \mu - 1, 5\mu + 2)$

$$\Rightarrow \vec{PQ} = (-3 - 2)\hat{i} + (-3)\hat{j} + (5 - 4)\hat{k}$$

$$\Rightarrow \vec{PQ} \cdot \hat{n} = 0 \text{ (where } \hat{n} \text{ is } \perp^{\text{er}} \text{ to plane)}$$

$$\Rightarrow (-3\mu - 2)1 + (\mu - 3) \cdot (-4) + (5\mu - 4)3 = 0$$

$$\Rightarrow \frac{1}{4}$$

Q44. If the distance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = a$, where $a > 0$, is 5, then the foot of the perpendicular from P to the plane is

$$(A) \left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3} \right) \quad (B) \left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3} \right) \quad (C) \left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3} \right) \quad (D) \left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2} \right)$$

Ans. A

$$\text{Sol. } \because \left| \frac{1 - 4 - 2 - a}{3} \right| = 5$$

$$\Rightarrow a = 10, -20$$

$$\Rightarrow \alpha = 10 \quad \because \alpha > 0$$

Now, let $Q(\alpha, \beta, \gamma)$ be the foot of perpendicular from P to the plane $x + 2y - 2z = 10$

$$\text{Equation of line PQ is } \frac{x-1}{1} = \frac{y+2}{2} = \frac{z-1}{-2} = r \quad (\text{Let})$$

$$\Rightarrow \alpha = r + 1, \beta = 2r - 2 \text{ and } \gamma = -2r + 1$$

$\because$ Q lies in the plane

$$\therefore (r + 1) + 2(2r - 2) - 2(-2r + 1) = 10$$

$$\Rightarrow r = \frac{5}{3}$$

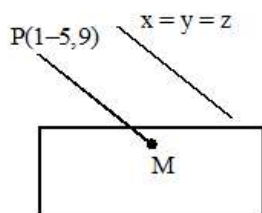
$$\text{foot of the perpendicular is } \left(\frac{8}{3}, \frac{4}{3}, \frac{7}{3} \right)$$

Q45. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is

$$(A) 3\sqrt{5} \quad (B) 10\sqrt{3} \quad (C) 5\sqrt{3} \quad (D) 3\sqrt{10}$$

Ans. B

Sol.



Equation of a line $\parallel$ to $x = y = z$ and passing through $(1, -5, 9)$ is

$$\frac{x-1}{1} = \frac{y+5}{1} = \frac{z-9}{1} = r$$

$$x - y + z = 5$$

$$r + 1 - r + 5 + r + 9 = 5$$

$$r = -10$$

$$\text{Hence } M(-9, -15, -1)$$

$$\text{Distance PM} = \sqrt{100 + 100 + 100} = 10\sqrt{3}$$

Q46. The equation of the plane containing the line $2x - 5y + z = 3$; $x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is

$$(A) 2x + 6y + 12z = 13 \quad (B) x + 3y + 6z = -7 \quad (C) x + 3y + 6z = 7$$

$$(D) 2x + 6y + 12z = -13$$

Ans. C

$$\begin{aligned} \text{Sol. } 2x - 5y + z - 3 + \lambda(x + y + 4z - 5) &= 0 \\ x(2 + \lambda) + y(\lambda - 5) + z(1 + 4\lambda) - (3 + 5\lambda) &= 0 \end{aligned}$$

$$\frac{2+\lambda}{1} = \frac{\lambda-5}{3} = \frac{1+4\lambda}{6}$$

$$\lambda = -\frac{11}{2}$$

So equation of plane
 $x + 3y + 6z = 7$

- Q47. The distance of the point $(1, 3, -7)$ from the plane passing through the point $(1, -1, -1)$ having normal perpendicular to both the lines $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-4}{3}$ and

$\frac{x-2}{2} = \frac{y+1}{-1} = \frac{z-7}{-1}$ is:-

- (A) $\frac{5}{\sqrt{83}}$ (B) $\frac{10}{\sqrt{74}}$ (C) $\frac{20}{\sqrt{74}}$ (D) $\frac{10}{\sqrt{83}}$

Ans. D

Sol.

$$\text{Normal vectro of plane} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & -1 & -1 \end{bmatrix}$$

$$= 5\hat{i} + 7\hat{j} + 3\hat{k}$$

equation of plane $5x + 7y + 3z = -5$

$$\text{Distance} = \left| \frac{5(1) + 7(3) + 3(-7) + 5}{\sqrt{83}} \right|$$

$$= \frac{10}{\sqrt{83}}$$

- Q48. If L_1 is the line of intersection of the planes $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ and L_2 is the line of intersection of the planes $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$, then the distance of the origin from the plane containing the lines L_1 and L_2 is

- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4\sqrt{2}}$ (D) $\frac{1}{3\sqrt{2}}$

Ans. D

Sol. required plane will pass though line of intersection of planes so equation of required plane will be

$$x + 2y - z - 3 + \lambda(3x - y + 2z - 1) = 0$$

take a fixed point on L_1 ($y = 0$)

$P(-5, 0, 4)$

point P will lie on required plane

$$\Rightarrow \lambda = -\frac{3}{2}$$

So equation of required plane will be

$$7x - 7y + 8z + 3 = 0$$

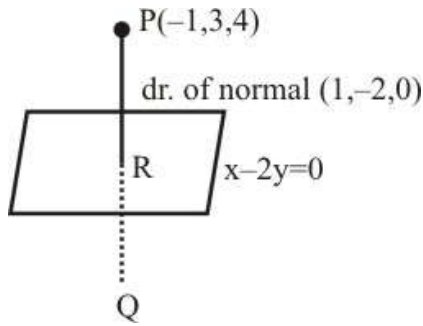
$$\text{distance from origin} = \left| \frac{3}{\sqrt{162}} \right| = \frac{1}{3\sqrt{2}}$$

- Q49. The image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is -

- (A) $(15, 11, 4)$ (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) $(8, 4, 4)$ (D) None of these

Ans. D

Sol.



Equation of line PQ is

$$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = \lambda$$

For some suitable value of λ , co-ordinates of point Q $(\lambda-1, -2\lambda, 4)$

R is the mid point of P and Q

$$\therefore R \equiv \left(\frac{\lambda-2}{2}, \frac{6-2\lambda}{2}, 4 \right)$$

$$R \equiv \left(\frac{\lambda-2}{2}, 3-\lambda, 4 \right)$$

It satisfies $x - 2y = 0$

$$\Rightarrow \lambda = \frac{14}{5}$$

$$\therefore Q = \left(\frac{2}{5}, \frac{1}{5}, 4 \right)$$

Q50. If the image of the point $P(1, -2, 3)$ in the plane, $2x + 3y - 4z + 22 = 0$ measured parallel to the line, $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q, then PQ is equal to

- (A) $\sqrt{42}$ (B) $6\sqrt{5}$ (C) $3\sqrt{5}$ (D) $2\sqrt{42}$

Ans. D

Sol. Equation of PQ

$$\vec{r} = (1, -2, 3) + \lambda(1, 4, 5)$$

$$N(1 + \lambda, 4\lambda - 2, 3 + 5\lambda)$$

$$2(1 + \lambda) + 3(4\lambda - 2) - 4(3 + 5\lambda) + 22 = 0$$

$$6\lambda = 6$$

$$N(2, 2, 8)$$

$$\Rightarrow Q(3, 6, 13)$$

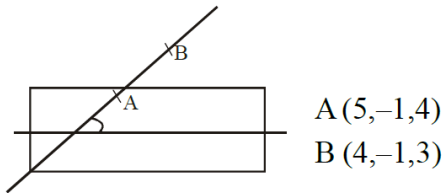
$$\Rightarrow PQ = 2\sqrt{42}$$

Q51. The length of the projection of the line segment joining the points $(5, -1, 4)$ and $(4, -1, 3)$ on the plane, $x + y + z = 7$ is

- (A) $\frac{1}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{2}{3}$

Ans. B

Sol.



θ = angle between plane & line

$$\cos(90 - \theta) = \frac{|\overline{AB} \cdot (\hat{i} + \hat{j} + \hat{k})|}{|\overline{AB}| |\hat{i} + \hat{j} + \hat{k}|} = \frac{2}{\sqrt{6}}$$

$$\text{Req. length} = |\overline{AB}| \cos \theta$$

$$= \sqrt{2} \times \frac{\sqrt{2}}{\sqrt{6}} = \sqrt{\frac{2}{3}}$$

Q52. Vector $\vec{x}$ satisfying the relation $\vec{A} \cdot \vec{x} = c$ and $\vec{A} \times \vec{x} = \vec{B}$ is

(A) $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|}$ (B) $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|^2}$ (C) $\frac{c\vec{A} + (\vec{A} \times \vec{B})}{|\vec{A}|^2}$ (D) $\frac{c\vec{A} - 2(\vec{A} \times \vec{B})}{|\vec{A}|^2}$

Ans. C

Sol. $\vec{A} \cdot \vec{x} = c$

$$\vec{A} \times \vec{x} = \vec{B}$$

X with $\vec{A}$

$$(\vec{A} \times \vec{x}) \times \vec{A} = \vec{B} \times \vec{A}$$

$$\vec{x}(\vec{A} \cdot \vec{A}) - \vec{A}(\vec{A} \cdot \vec{x}) = \vec{B} \times \vec{A}$$

$$\vec{x} = \frac{\vec{B} \times \vec{A} + c\vec{A}}{|\vec{A}|}$$

$$= \frac{c\vec{A}}{\vec{A} \times \vec{B}}$$

Q53. Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ and $\vec{a} \cdot \vec{b} = 3$ is

(A) $-\hat{i} + \hat{j} - 2\hat{k}$ (B) $2\hat{i} - \hat{j} + 2\hat{k}$ (C) $\hat{i} - \hat{j} - 2\hat{k}$ (D) $\hat{i} + \hat{j} - 2\hat{k}$

Ans. A

Sol. $(\vec{a} \times \vec{b}) + \vec{c} = \vec{0}$

$$(\vec{a} \times \vec{b}) = -\vec{c}$$

$$\Rightarrow \vec{a}(\vec{a} \times \vec{b}) = -\vec{a} \times \vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2\vec{b} = -\vec{a} \times \vec{c}$$

$$\Rightarrow 3(\vec{j} - \vec{k}) - 2\vec{b} = -(-2\vec{i} - \vec{j} - \vec{k})$$

$$(\vec{a} \times \vec{c} = 2\vec{i} - \vec{j} - \vec{k})$$

$$\Rightarrow 2\vec{b} = (-2\vec{i} + 2\vec{j} - 4\vec{k})$$

$$\Rightarrow \vec{b} = -\vec{i} + \vec{j} - 2\vec{k}$$

Q54. If $\vec{a}, \vec{b}, \vec{c}$ are linearly independent vectors, then which one of the following set of vectors is linearly dependent?

(A) $\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}$ (B) $\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}$ (C) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ (D) None

Ans. B

Sol. $\because \vec{a}, \vec{b}, \vec{c}$ are linearly independent then $[\vec{a}\vec{b}\vec{c}] \neq 0$

If there vector $\vec{p}, \vec{q}, \vec{r}$ are linearly dependent vector then $[\vec{p}\vec{q}\vec{r}] = 0$

(A) $[\vec{a} - \vec{b} \quad \vec{b} - \vec{c} \quad \vec{c} - \vec{a}] = 0$

(C) $[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a}\vec{b}\vec{c}]^2 \neq 0$

Exercise J02

Q1. If $\vec{a}$ & $\vec{b}$ are non collinear vectors such that, $\vec{p}(x+4y)\vec{a}+(2x+y+1)\vec{b}$ & $\vec{q}(y-2x+2)\vec{a}+(2x-3y-1)\vec{b}$, find x & y such that $3\vec{p} = 2\vec{q}$.

Ans. $x = 2, y = -1$

Sol. $\vec{a} \nparallel \vec{b}$

$$3\vec{p} = 2\vec{q}$$

$$3((x+4y)\vec{a} + (2x+y+1)\vec{b}) = 2((y-2x+2)\vec{a} + (2x+3y-1)\vec{b})$$

$$(3x+12y-2y+4x-4)\vec{a} = (4x-6y-2-6x-3y)\vec{b}$$

$$(7x+10y-4)\vec{a} = (-2x-9y-5)\vec{b}$$

$$7x+10y-4=0 \quad \dots\dots\dots(1)$$

$$-2x-9y-5=0 \quad \dots\dots\dots(2)$$

from (1) & (2)

$$x = 2, y = -1$$

Q2. Show that the points $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ are collinear.

Ans. -

Sol. $\left. \begin{array}{l} \vec{p} = \vec{a} - 2\vec{b} + 3\vec{c} \\ \vec{q} = 2\vec{a} + 3\vec{b} - 4\vec{c} \\ \vec{r} = -7\vec{b} + 10\vec{c} \end{array} \right\} \text{Collinear}$

$$2\vec{p} - \vec{q} = \vec{r}$$

$$\vec{qp} = -\vec{a} - 5\vec{b} + 7\vec{c}$$

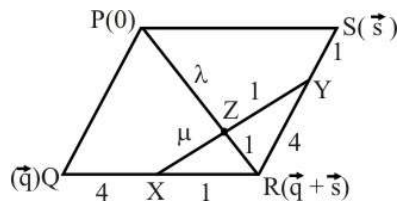
$$\vec{rq} = 2\vec{a} + 10\vec{b} + 14\vec{c}$$

$$\vec{qp} = \lambda \vec{rq}$$

Q3. Points X & Y are taken on the sides QR & RS, respectively of a parallelogram PQRS, so that $\vec{QX} = 4\vec{XR}$ & $\vec{RY} = 4\vec{YS}$. The line XY cuts the line PR at Z. Prove that $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

Ans. -

Sol.



$$\overrightarrow{QX} = 4\overrightarrow{XR}$$

$$\overrightarrow{RY} = 4\overrightarrow{YS}$$

Let $\vec{P}$ be origin & R $(\vec{q} + \vec{s})$ from figure

$$\text{P.V. of X} = \frac{4(\vec{q} + \vec{s}) + \vec{q}}{5} = \frac{5\vec{q} + 4\vec{s}}{5}$$

$$\text{P. V. of Y} = \frac{4\vec{s} + \vec{q} + \vec{s}}{5} = \frac{5\vec{s} + \vec{q}}{5}$$

Now Let Z divides PR in ratio $\lambda : 1$

Now Let Z divides XY in ratio $\mu : 1$

$$\text{P.V. of Z} = \frac{\lambda(\vec{q} + \vec{s})}{\lambda + 1} \quad (\text{from PR})$$

$$\text{P.V. of Z} = \frac{\frac{\mu(5\vec{s} + \vec{q})}{5} + \frac{5\vec{q} + 4\vec{s}}{5}}{\mu + 1} \quad (\text{from XY})$$

equating both Z then we get

$$\frac{\lambda}{\lambda + 1} = \frac{\mu + 5}{5(\mu + 1)} \quad \dots\dots\dots(i)$$

$$\frac{\lambda}{\lambda + 1} = \frac{5\mu + 4}{5(\mu + 1)} \quad \dots\dots\dots(ii)$$

$$\text{from (i) \& (ii), } \mu = \frac{1}{4} \text{ \& } \lambda = \frac{21}{4}$$

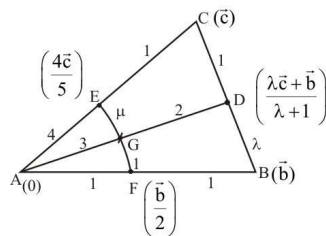
$$\text{So P.V. of Z} = \frac{\frac{21}{4}}{\frac{21}{4} + 1}$$

$$= \frac{21}{25}(\vec{q} + \vec{s}) = \frac{21}{25}\overrightarrow{PR}$$

- Q4. In a ΔABC , points E and F divide sides AC and AB respectively so that $\frac{AE}{EC} = 4$ and $\frac{AR}{FB} = 1$. Suppose D is a point on side BC. Let G be the intersection of EF and AD and suppose D is situated so that $\frac{AG}{GD} = \frac{3}{2}$. If the ratio $\frac{BD}{DC} = \frac{a}{b}$, where a and b are in their lowest form, find the value of (a + b).

Ans. 9

Sol.



$$\text{Let } \frac{BD}{CD} = \frac{\lambda}{1}$$

PV. of G w.r.t. $\in$ F

$$\text{Let } \frac{EG}{GF} = \mu$$

$$\frac{\frac{3(\lambda+b)}{\lambda+1}}{5} = \frac{5\mu b + 8c}{10(\mu+1)}$$

$$\frac{3\lambda}{5(\lambda+1)} = \frac{8}{10(\mu+1)} \dots\dots\dots(1), \quad \frac{3}{5(\lambda+1)} = \frac{5\mu}{10(\mu+1)} \dots\dots\dots(1)$$

$$\lambda = \frac{4}{5}, \quad \mu = 2$$

$$\frac{BD}{CD} = \frac{4}{5}$$

Q5. Find a unit vector $\hat{a}$ which makes an angle $(\pi/4)$ with axis of z & is such that $\hat{a} + \hat{i} + \hat{j}$ is a unit vector.

Ans. $\frac{-1}{2} \hat{i} - \frac{1}{2} \hat{j} + \frac{1}{\sqrt{2}} \hat{k}$

Sol. Let $\hat{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$

$$\hat{a} \cdot \hat{k} = \cos \frac{\pi}{4}$$

$$a_3 = \frac{1}{\sqrt{2}}$$

$$a_1^2 + a_2^2 = \frac{1}{2}$$

$$\hat{a} + \hat{i} + \hat{j} = (a_1 + 1)\hat{i} + (a_2 + 1)\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$(a_1 + 1)^2 + (a_2 + 1)^2 = \frac{1}{2}$$

$$2a_1 + 1 + 2a_2 + 1 = 0$$

$$a_1 + a_2 = -1 \quad \dots(1)$$

$$2a_1 a_2 = +\frac{1}{2}$$

$$a_1 a_2 = \frac{1}{4} \quad \dots\dots(2)$$

from (1) & (2)

$$a_1 = \frac{-1}{2}, \quad a_2 = -\frac{1}{2}$$

Q6. Prove that $\left(\frac{\vec{a}}{|\vec{a}|^2} - \frac{\vec{b}}{|\vec{b}|^2} \right) = \left(\frac{\vec{a} - \vec{b}}{|\vec{a}| |\vec{b}|} \right)^2$

Ans. -

Sol.

$$\begin{aligned} \left(\frac{\vec{a}}{(\vec{a})^2} - \frac{\vec{b}}{(\vec{b})^2} \right) &= \left(\frac{\vec{a} |\vec{b}| - \vec{b} |\vec{a}|^2}{|\vec{a}|^2 \cdot |\vec{b}|^2} \right)^2 \\ &= \left(\frac{\vec{a} \cdot \vec{b} (\vec{b} - \vec{a})}{(\vec{a} \cdot \vec{b})^2} \right)^2 \\ &= \frac{\vec{b} - \vec{a}}{\vec{a}} \end{aligned}$$

Q7. If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then find the range of $\frac{3|\vec{a} + \vec{b}|}{2} + 2|\vec{a} - \vec{b}|$.

Ans. Range: [3, 5]

Sol.

$$\begin{aligned} \frac{3}{2} |\vec{a} + \vec{b}| + 2 |\vec{a} - \vec{b}| \quad & \quad |\vec{a} + \vec{b}| = 2 \cos \frac{\alpha}{2} \\ & \quad |\vec{a} - \vec{b}| = 2 \sin \frac{\alpha}{2} \end{aligned}$$

$$3 \cos \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2}$$

[3, 5]

Q8. Let $\vec{V}_1$ and $\vec{V}_2$ are two vectors such that $\vec{V}_1 = 2(\sin \alpha + \cos \alpha) \hat{i} + \hat{j}$ and $\vec{V}_2 = \sin \beta \hat{i} + \cos \beta \hat{j}$ where α and β satisfy the relation $2(\sin \alpha + \cos \alpha) \sin \beta = 3 - \cos \beta$, find the value of $(3 \tan^2 \alpha + 4 \tan^2 \beta)$.

Ans. 35

Sol. We have $\vec{V}_1 \cdot \vec{V}_2 = 2(\sin \alpha + \cos \alpha) \sin \beta + \cos \beta$

$\therefore \vec{V}_1 \cdot \vec{V}_2 = 3$ [Given that $2(\sin \alpha + \cos \alpha) \sin \beta + \cos \beta = 3$]

$|\vec{V}_1|^2 |\vec{V}_2|^2 \cos^2 \theta = 9 \quad (\theta = \vec{V}_1 \wedge \vec{V}_2)$

$$[4(1 + \sin 2\alpha) + 1](1) \cos^2 \theta = 9$$

$$\cos^2 \theta = \frac{9}{5 + 4\sin 2\alpha} \leq 1 \Rightarrow 9 \leq 5 + 4\sin 2\alpha \Rightarrow 4 \leq 4\sin 2\alpha$$

$$\therefore \sin 2\alpha = 1 \Rightarrow \alpha = \frac{\pi}{4}$$

$$\therefore \cos^2 \theta = \frac{9}{9} = 1 \Rightarrow \theta = 0$$

$$\text{Hence } \theta = 0 \Rightarrow \vec{V}_1 \text{ and } \vec{V}_2 \text{ are collinear i.e. } = \frac{2(\sin \alpha + \cos \alpha)}{\sin \beta} = \frac{1}{\cos \beta} \quad \left(\alpha = \frac{\pi}{4} \right)$$

$$\tan \beta = 2 \cdot 2 \left(\frac{2}{\sqrt{2}} \right) = 2\sqrt{2}$$

$$\text{Hence } 3 \tan^2 \alpha + 4 \tan^2 \beta = 3 + (4)(8) = 35$$

Q9. In a unit cube. Find

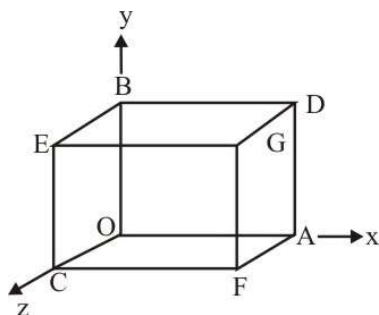
(a) The angle between the diagonal of the cube and a diagonal of a face skew to it.

(b) The angle between the diagonals of two faces of the cube through the same vertex.

(c) The angle between a diagonal of a cube and a diagonal of a face intersecting it.

Ans. (a) $\cot^{-1}(0)$; (b) $\cot^{-1} \frac{1}{\sqrt{3}}$; (c) $\cot^{-1} \sqrt{2}$

Sol.



$$O(0,0,0) \quad A(1,0,0), \quad B(0,1,0)$$

$$C(0,0,1) \quad D(1,1,0) \quad E(0,1,1)$$

$$F(1,0,1) \quad G(1,1,1)$$

$$(a) \text{ Cube diagonal } \vec{OG} = (1,1,1)$$

$$\text{face diagonal } \vec{BC} = (0, -1, 1)$$

$$\cos \theta = \frac{0 - 1 + 1}{\sqrt{3}\sqrt{2}} = 0$$

$$\theta = \frac{\pi}{2}$$

$$(b) \vec{OF} = (1, 0, 1), \quad \vec{OD} = (1, 1, 0)$$

$$\cos \theta = \frac{\vec{OF} \cdot \vec{OD}}{|\vec{OF}| |\vec{OD}|} = \frac{1 + 0 + 0}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$\vec{OG} = (1, 1, 1)$$

$$\vec{OP} = (1, 1, 0)$$

$$\cos \theta = \frac{1 + 1 + 0}{\sqrt{2} \cdot \sqrt{2}} = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\cot \theta = \frac{\sqrt{2}}{1}$$

$$\theta = \cot^{-1}(\sqrt{2})$$

Q10. Given that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{v} = 2\hat{i} + \hat{j} + 4\hat{k}$; $\vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ and $(\vec{u} \cdot \vec{R} - 10)\hat{i} + (\vec{v} \cdot \vec{R} - 20)\hat{j} + (\vec{w} \cdot \vec{R} - 20)\hat{k} = \vec{0}$. Find the unknown vector $\vec{R}$.

Ans. $-\hat{i} + 2\hat{j} + 5\hat{k}$

Sol. $(\vec{u} \cdot \vec{R} - 10)\hat{i} + (\vec{v} \cdot \vec{R} - 20)\hat{j} + (\vec{w} \cdot \vec{R} - 20)\hat{k} = \vec{0}$

$$\vec{u} \cdot \vec{R} - 10 = 0$$

$$\vec{u} \cdot \vec{R} = 10$$

$$R_1 - 2R_2 + 3R_3 = 10 \quad \dots\dots\dots(1)$$

$$\left| \begin{array}{l} \vec{v} \cdot \vec{R} = 0 \\ 2R_1 + R_2 + 4R_3 = 20 \end{array} \right| \quad \dots\dots\dots(2)$$

$$\vec{w} \cdot \vec{R} = 20$$

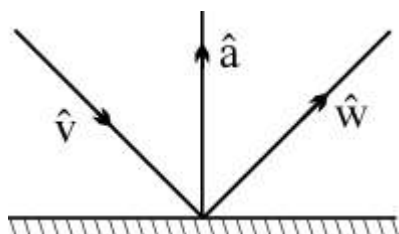
$$R_1 + 3R_2 + 3R_3 = 20 \quad \dots\dots\dots(3)$$

from (1), (2) & (3)

$$R_1 = -1, R_2 = , R_3 = 5$$

$$\vec{R} = (-\hat{i} + 2\hat{j} + 5\hat{k})$$

Q11. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.



Ans. $\hat{w} = \hat{v} - 2(\hat{a} \cdot \hat{v})\hat{a}$

Sol. Given that incident ray is along $\hat{v}$, reflected ray is along $\hat{w}$ normal $\hat{a}$, outwards. The given figure can be redrawn as shown.

We know that incident ray, reflected ray and normal lie in a plane, and angle of incidence = angle of reflection.

Therefore $\hat{a}$ will be along the angle bisector of $\hat{w}$ and $-\hat{v}$, i.e.,

$$\hat{a} = \frac{\hat{w} + (-\hat{v})}{|\hat{w} - \hat{v}|} \dots\dots\dots(i)$$

[$\therefore$ angle bisector will along a vector dividing in same ratio as the ratio of the sides forming that angle.]

But $\hat{a}$ is a unit vector

where $|\hat{w} - \hat{v}| = OC = 2OP$

$$= 2|\hat{w}|\cos\theta = 2\cos\theta$$

Substituting this value in equation (i) we get

$$\hat{a} = \frac{\hat{w} - \hat{v}}{2\cos\theta}$$

$$\therefore \hat{w} = \hat{v} + (2\cos\theta)\hat{a}$$

$$= \hat{v} - 2(\hat{a} \cdot \hat{v})\hat{a} \quad [\because \hat{a} \cdot \hat{v} = -\cos\theta]$$

Q12. Find the minimum area of the triangle whose vertices are A(-1, 1, 2); B(1, 2, 3) and C(t, 1, 1) where t is a real number.

Ans. $\frac{\sqrt{3}}{2}$

Sol. (a) $\overrightarrow{AB}(2, 1, 1)$

$$\overrightarrow{AC}(t+1, 0, -1)$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ t+1 & 0 & -1 \end{vmatrix}$$

$$= \hat{i}(-1-0) - \hat{j}(-2-t-1) + \hat{k}(t+1)$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = -\hat{i} + (3+t)\hat{j} + (t+1)\hat{k}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}|^2 = 1 + (t+3)^2 + (t+1)^2$$

$$= 1 + t^2 + 9 + 6t + t^2 + 2t + 1$$

$$= 2t^2 + 8t + 11$$

$$= 2(t^2 + 4t) + 11$$

$$|\vec{AB} \times \vec{AC}|^2 = 2(t+2)^2 + 3$$

$$A = \frac{1}{2} \sqrt{2(t+2)^2 + 3}$$

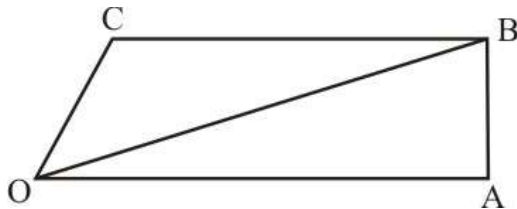
$$A_{\min} = \frac{1}{2} \sqrt{3} = \frac{\sqrt{3}}{2}$$

- Q13. Let $\vec{OA} = \vec{a}$; $\vec{OB} = 100\vec{a} + 2\vec{b}$ and $\vec{OC} = \vec{b}$ where O, A, and C are non collinear points. Let P denotes the area of the parallelogram with $\vec{OA}$ and $\vec{OC}$ as adjacent sides and Q denotes the area of the quadrilateral OABC. If $Q = \lambda P$. Find the value of λ .

Ans. 51

Sol. $P = |\vec{OA} \times \vec{OC}|$

$$P = |\vec{a} \times \vec{b}|$$



$$(Q) \text{ Area OABC} = \Delta OAB + \Delta OBC$$

$$= \frac{1}{2} |\vec{OA} \times \vec{OB}| + \frac{1}{2} |\vec{OB} \times \vec{OC}|$$

$$= \frac{1}{2} |2\vec{a} \times \vec{b}| + \frac{1}{2} |100\vec{a} \times \vec{b}|$$

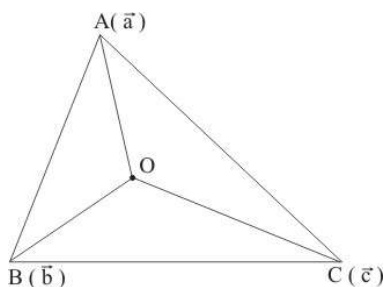
$$= 51 |\vec{a} \times \vec{b}|$$

$$Q = 51 P$$

- Q14. Let O be the interior point of ΔABC such that $2\vec{OA} + 3\vec{OB} + 6\vec{OC} = \vec{0}$. If $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ where m and n are relatively prime, then find (m + n).

Ans. 17

Sol.



$$\text{Area of } \Delta ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

Now given $2\vec{a} + 3\vec{b} + 6\vec{c} = 0$

Cross with $\vec{a}$, $3\vec{a} \times \vec{b} + 6\vec{a} \times \vec{c} = 0$

$$\text{or } \vec{a} \times \vec{b} = 2(\vec{c} \times \vec{a}) = 0 \dots\dots\dots(1)$$

again cross with $\vec{b}$, $2\vec{a} \times \vec{b} + 6\vec{c} \times \vec{b} = 0$

$$\text{or } \vec{a} \times \vec{b} = (\vec{b} \times \vec{c}) = 0 \dots\dots\dots(2)$$

$$\text{Area of } \Delta OAB = \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$\text{Ar. of } \frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{\frac{1}{2} |\vec{a} \times \vec{b}| (1 + \frac{1}{2} + \frac{1}{3})}{\frac{1}{2} |\vec{a} \times \vec{b}|} = \frac{11}{3} \Rightarrow m + n = 17$$

Q15. Determine vector of magnitude 9 which is perpendicular to both the vectors:

$$4\hat{i} - \hat{j} + 3\hat{k} \text{ \& } -2\hat{i} - \hat{j} - 2\hat{k}$$

$$\text{Ans. } \pm 3 (\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\begin{aligned} \text{Sol. } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} &= \hat{i}(-1) - \hat{j}(-2) + \hat{k}(+2) \\ &= (-1, +2, 2) \end{aligned}$$

$$\vec{a} = \pm 3(-1, 2, 2)$$

Q16. A triangle has vertices (1, 1, 1) ; (2, 2, 2), (1, 1, y) and has the area equal to cosec ($\pi/4$)sq. units.

Find the value of y.

$$\text{Ans. } y = 3 \text{ or } y = -1$$

$$\begin{aligned} \text{Sol. } \text{Arg} &= \sqrt{2} \\ \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 2y \end{vmatrix} &= 2\sqrt{2} \\ \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 1-y \end{vmatrix} & \end{aligned}$$

$$\left| (1-y) (\hat{i} - \hat{j}) \right| = 2\sqrt{2}$$

$$(1-y)\sqrt{2} = \pm 2\sqrt{2}$$

$$y = 1 \pm 2$$

$$= 3, -1$$

Q17. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1} \left(\frac{1}{4} \right)$.

If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = 4$, $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$ and $\vec{c} = \lambda \vec{a} + \mu \vec{b}$, then, find

(a) the value of λ , (b) the sum of values of μ and (c) the product of all possible values of μ .

Ans. (a) 2, (b) -1, (c) -12

Sol. $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$

$$(\vec{c} - 2\vec{a}) \times \vec{b} = 0$$

$$\vec{c} = 2\vec{a} + \lambda \vec{b}$$

$$|\vec{c}|^2 = 4a^2 + \lambda^2 b^2 + 4\lambda \vec{a} \cdot \vec{b}$$

$$16 = 4 + \lambda^2 + 4\lambda \left(\frac{1}{4} \right)$$

$$\lambda^2 + \lambda - 12 = 0$$

$$\lambda = 3, \lambda = -4$$

$$\vec{c} = 2\vec{a} + 3\vec{b}$$

$$\vec{c} = 2\vec{a} - 4\vec{b}$$

$$\vec{c} = 2\vec{a} - 4\vec{b}$$

$$\lambda = 2, \mu = 3, \mu = -4$$

Q18. Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four distinct vectors satisfying $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$. Show that

$$\vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d} \neq \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{d}$$

Ans. -

Sol. Given that $\vec{a} \neq \vec{b} \neq \vec{d}$

Such that $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ (i)

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{d} \text{(ii)}$$

To prove that $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) \neq 0$

Subtracting equation (ii) from (i) we get

$$\vec{a} \times (\vec{c} \times \vec{b}) = (\vec{b} - \vec{c}) \times \vec{d}$$

$$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = \vec{d} \times (\vec{c} - \vec{b}) = 0$$

$$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) - \vec{d} \times (\vec{c} - \vec{b}) = 0$$

$$\Rightarrow (\vec{a} - \vec{d}) \times (\vec{c} - \vec{b}) = 0 \Rightarrow \vec{a} - \vec{d} \parallel \vec{c} - \vec{b}$$

$[\because \vec{a} - \vec{d} \neq 0 \text{ as all distinct}]$

$\Rightarrow$ Angle between $\vec{a} - \vec{d}$ and $\vec{c} - \vec{b}$ is either 0 or 180.

$$\Rightarrow (\vec{a} - \vec{d}) \cdot (\vec{c} - \vec{b}) = |\vec{a} - \vec{d}| |\vec{c} - \vec{b}|$$

$[\cos 0 \text{ or } \cos 180] \neq 0 \text{ as } \vec{a}, \vec{d}, \vec{c}, \vec{b} \text{ all are different.}$

Q19. Let $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

Ans. 9

Sol. (c) $\vec{a} = -\hat{i} - \hat{k}$

$$\vec{b} = -\hat{i} + \hat{j}$$

$$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$$

Taking cross product by $\vec{a}$

$$(\vec{r} \times \vec{b}) \times \vec{a} = (\vec{c} \times \vec{b}) \times \vec{a}$$

$$\Rightarrow (\vec{r} \cdot \vec{a})\vec{b} - (\vec{b} \cdot \vec{a})\vec{r} = (\vec{c} \cdot \vec{a})\vec{b} - (\vec{b} \cdot \vec{a})\vec{c}$$

$$\Rightarrow 0 - = -\vec{r} = (1 - 1 - 3) - (-\hat{i} + \hat{j}) - (1)(\hat{i} + 2\hat{j} + 3\hat{k})$$

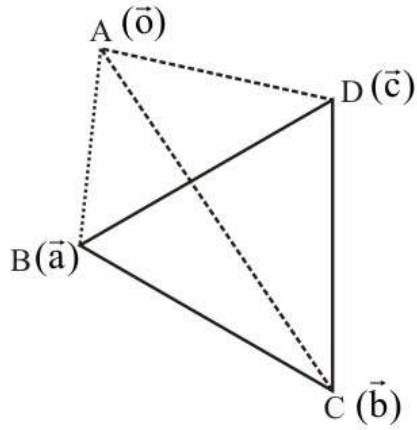
$$\vec{r} = -3\hat{i} + 6\hat{j} + 3\hat{k}$$

$$\vec{r} \cdot \vec{b} = 3 + 6 = 9$$

Q20. Let the volume of a tetrahedron having the vertices as $\vec{0}, \vec{a}, \vec{b}$ & $\vec{c}$ is λ_1 (unit cubes) & volume of another tetrahedron formed by the centroids of all faces of given tetrahedron is λ_2 (in unit cubes), then $\frac{\lambda_1}{3\lambda_2}$ is

Ans. 9

Sol.



$$\lambda = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$$

$$G_1 (A, B, C) = \left(\frac{\vec{a} + \vec{b}}{3} \right)$$

$$G_2 (ABD) = \left(\frac{\vec{a} + \vec{c}}{3} \right)$$

$$G_3 (BCD) = \left(\frac{\vec{a} + \vec{b} + \vec{c}}{3} \right)$$

$$G_4 (ACD) = \left(\frac{\vec{b} + \vec{c}}{3} \right)$$

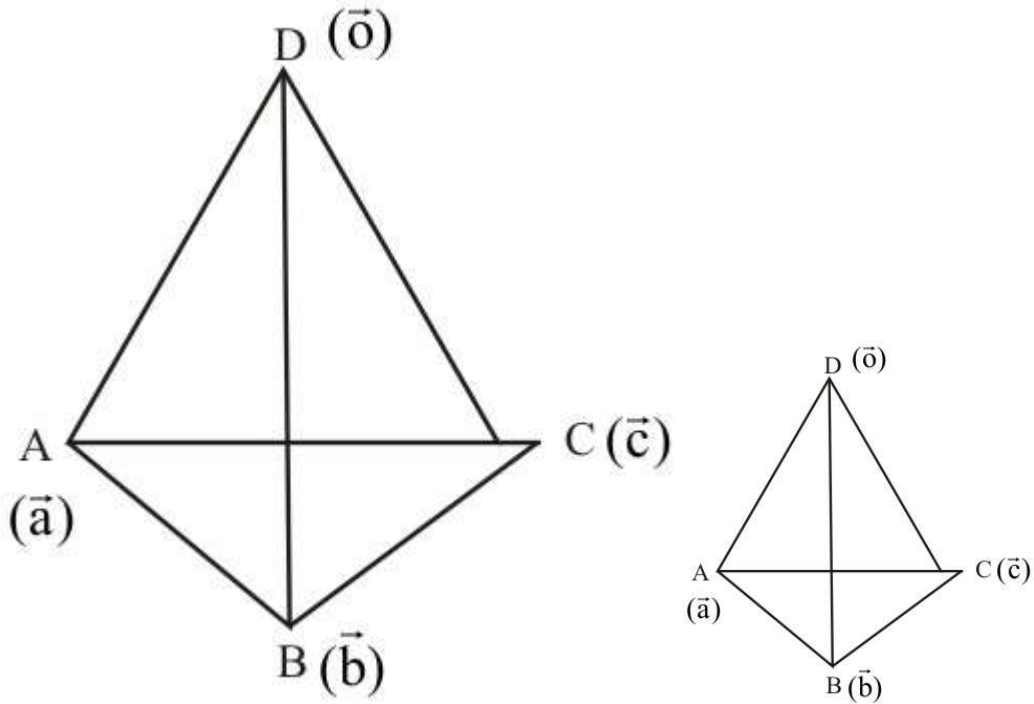
$$\overrightarrow{G_3 G_1} = \frac{\vec{c}}{3}, \quad \overrightarrow{G_3 G_2} = \frac{\vec{b}}{3}, \quad \overrightarrow{G_3 G_4} = \frac{\vec{a}}{3}$$

$$\lambda_2 = \frac{1}{6} \left[\frac{\vec{a} + \vec{b} + \vec{c}}{3} \right] = \frac{1}{27} \lambda_1$$

Q21. Let the volume of tetrahedron ABCD is 81 cubic units & G_1, G_2, G_3 are centroids of triangular faces ABC, ABD & ACD respectively, then volume of tetrahedron $AG_1 G_2 G_3$, is (in cubic units)

Ans. 6

Sol.



$$= \frac{1}{6} [\vec{AD} \cdot \vec{BD} \cdot \vec{CD}]$$

$$\Rightarrow \frac{1}{6} [\vec{abc}] = 81$$

$$G_1 = \frac{\vec{a} + \vec{b} + \vec{c}}{3}, G_2 = \frac{\vec{a} + \vec{b}}{3}, G_3 = \frac{\vec{b} + \vec{c}}{3}$$

$$A(\vec{a})$$

$$V = \frac{1}{6} [G_1 G_2 G_1 G_3 G_1 A]$$

$$= \frac{1}{6} \cdot \frac{1}{27} (\vec{c} \times \vec{b}) \cdot (\vec{b} + \vec{c} - 2\vec{a})$$

$$= \frac{1}{6} \cdot \frac{1}{27} 2[\vec{abc}]$$

$$\frac{2}{27} \times 81$$

$$= 6$$

Q22. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$, then the value of $(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$, is

Ans. 5

$$\begin{aligned} \text{Sol. } & (2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times \vec{a} - 2(\vec{a} \times \vec{b}) \times \vec{b}] \\ &= (2\vec{a} + \vec{b}) \cdot [a^2 \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a} - 2(\vec{a} \cdot \vec{b}) \vec{b} - b^2 \vec{a}] \\ &= (2\vec{a} + \vec{b}) \cdot [2\vec{a} + \vec{b}] \text{ as } [a^2 = b^2 = 1] \end{aligned}$$

$$\Rightarrow 4a^2 + b^2 = 5$$

Q23. Find the angle between the two straight lines whose direction cosines l, m, n are given by

$$2l + 2m - n = 0 \quad \text{and} \quad mn + nl + lm = 0.$$

Ans. $\theta = 90^\circ$

Sol. $n = (l + m) \dots\dots\dots(1)$

$$n(l + m) + l + m = 0 \dots\dots\dots(2)$$

$$2(l + m) + l + m = 0$$

$$2l^2 + 2m^2 + 5lm = 0$$

$$2l^2 + lm + 4lm + 2m^2 = 0$$

$$l(2l + m) + 2m(2 + lm) = 0$$

$$(l + 2m)(2l + m) = 0$$

$$l + 2m = 0$$

$$l = -2m$$

$$n = -2m$$

$$2l + m = 0$$

$$l = \frac{-m}{2}$$

$$n = m$$

$$(l_1, m_1, n_1) = (-2m, m, -2m) \quad (l_2, m_2, n_2) = \left(\frac{-m}{2}, m, m\right)$$

$$\cos \theta = \frac{(-2m)\left(\frac{-m}{2}\right) + m^2 - 2m^2}{\sqrt{4m^2 + m^2 + 4m^2} \sqrt{\frac{m^2}{4} + m^2 + m^2}}$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

Q24. Find out whether the following pairs of lines are parallel, non-parallel & intersecting, or non-parallel & non-intersecting.

$$\begin{aligned} \text{(i)} \quad \vec{r}_1 &= \hat{i} + \hat{j} + 2\hat{k} + \lambda(3\hat{i} - 2\hat{j} + 4\hat{k}) \\ \vec{r}_2 &= 2\hat{i} + \hat{j} + 3\hat{k} + \mu(-6\hat{i} + 4\hat{j} - 8\hat{k}) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \vec{r}_1 &= \hat{i} - \hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k}) \\ \vec{r}_2 &= 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu(2\hat{i} + \hat{j} + 3\hat{k}) \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \vec{r}_1 &= \hat{i} + \hat{k} + \hat{k} + \lambda(\hat{i} + 3\hat{j} + 4\hat{k}) \\ \vec{r}_2 &= 2\hat{i} + \hat{k} + 3\hat{k} + \mu(4\hat{i} - \hat{j} + \hat{k}) \end{aligned}$$

Ans. (i) parallel (ii) the lines intersect at the point p.v. $-2\hat{i} + 2\hat{j}$ (iii) lines are skew

Sol. (i) $\vec{r}_1 = (\hat{i} + \hat{j} + 2\hat{k})\lambda(3\hat{i} - 2\hat{j} + 4\hat{k}) \dots\dots\dots L_1$

$$(\vec{a}_1) + \lambda \vec{P}_1$$

$$\vec{r}_2 = (2\hat{i} + \hat{j} + 3\hat{k}) + \mu(-6\hat{j} + 4\hat{j} + 3\hat{k}) \dots\dots\dots L_2$$

$$\vec{a}_2 + \mu \vec{P}_2$$

$$\therefore \vec{P}_2 = -2\vec{P}_1$$

Hence $L_1 \parallel L_2$

(ii) $\vec{r}_1 = (\hat{i} - \hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}) \dots\dots\dots L_1$

$$\vec{r}_1 = \vec{a} + \lambda \vec{P}_1$$

$$\vec{r}_2 = 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu(2\hat{i} + \hat{j} + 3\hat{k}) \dots\dots\dots L_2$$

$$\vec{a}_2 + \mu \vec{P}_2$$

$$\vec{P}_1 \neq t \vec{P}_2 \text{ (Not parallel)}$$

Let Point on L_1 P $(1 + \lambda, -1 - \lambda, 3 + \lambda)$

Let point on L_2 O $(2 + 2\mu, 4 + -4, 6 + 3\mu)$

if L_1 & L_2 intersecting

$$1 + \lambda = 2 + 2\mu$$

$$\lambda - 2\mu = 1 \dots\dots\dots (1)$$

$$-1 - \lambda = 4 + \mu$$

$$\lambda + \mu = -5 \dots\dots\dots (2)$$

$$3 + \lambda = 6 + 3\mu$$

$$\lambda - 3\mu = 3 \dots\dots\dots (3)$$

from (1) & (2)

$$\mu = -2$$

$$\lambda = -3$$

putting in (3)

$$-3 - 3(2) = 3 \text{ satisfic}$$

Hence line are intersetting.

(iii) Let point on L_1 P $(1 + \lambda, 3\lambda, 1 + 4\lambda)$

Let point on L_1 O $(2 + 4\mu, 3 - \mu, \mu)$

let line are intersecting

then $1 + \lambda = 2 + 4\mu$

$$\lambda - 4\mu = 1 \dots\dots\dots(1)$$

$$3\lambda = -3\mu$$

$$3\lambda + \mu = 3 \dots\dots\dots(2)$$

$$1 + \mu\lambda = \mu$$

$$4\lambda = \mu = -1 \dots\dots\dots(3)$$

From (2) & (3)

$$7\lambda = 2$$

$$\lambda = \frac{2}{7}, \quad \mu = \frac{15}{7}$$

Put in (1)

$$\frac{2}{7} - \frac{60}{7} \neq 1 \text{ Hence line are skew.}$$

Q25. Find the equations of the straight line passing through the point $(1, 2, 3)$ to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z+4}{2} = \lambda \dots\dots\dots L_1$

let required line intersect L_1 at point P

$$P(2\lambda - 1, \lambda + 2, 2\lambda - 4)$$

Required line passing through Q $(1, 2, 3)$

Direction vector of line is $\overrightarrow{PQ}$

$$\overrightarrow{PQ} = (2\lambda - 2, \lambda, 2\lambda - 7)$$

line parallel to given plane

$$\overrightarrow{pa} \cdot \vec{n} = 0$$

$$(2\lambda - 2)(1) + \lambda(5) + (2\lambda - 7)(4) = 0$$

$$2\lambda - 2 + 5\lambda + 8\lambda - 28 = 0$$

$$15\lambda = 30$$

$$\lambda = 2$$

$$\overrightarrow{PQ} = (2, 2, -3)$$

$$\vec{r} = (1, 2, 3) + \lambda(2, 2, -3)$$

Q26. Find the equations of the two lines through the origin which intersect the line

$$\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1} \text{ at an angle of } \frac{\pi}{3}.$$

Ans. $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ or $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

Sol. Let point on line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{1}$ (1)

are $(3 + 2\lambda, 3 + \lambda, \lambda)$

Equation on line which pass through origin is

$$\frac{x-0}{3+2\lambda} = \frac{y-0}{3+\lambda} = \frac{z-0}{\lambda} \text{(2)}$$

Angle between (1) & (2)

$$\cos \frac{\pi}{3} = \frac{(3+2\lambda)2 + (3+\lambda)1 + \lambda \times 1}{\sqrt{(3+2)^2 + (3+\lambda)^2 + \lambda^2} \sqrt{2^2 + 1^2 + 1}}$$

Solving we get

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\Rightarrow \lambda = -1, -2$$

Putting the value of λ in equation (2)

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1} \text{ or } \frac{x}{-1} = \frac{y}{1} = \frac{z}{2}$$

Q27. Find the equation to the line passing through the point $(1, -2, -3)$ and parallel to the line $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$.

Ans. $\frac{x-1}{6} = \frac{y+2}{13} = \frac{z+3}{17}$

Sol. Direction vector of Req. line is

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & -3 \\ 3 & -4 & 2 \end{vmatrix}$$

$$= \hat{i}(6 - 12) - \hat{j}(4 + 9) + \hat{k}(-8 - 9)$$

$$= -6\hat{i} - 13\hat{j} - 17\hat{k}$$

$$\text{Equation } \frac{x-1}{6} = \frac{y+2}{13} = \frac{z+3}{17} = \lambda$$

Q28. Find the equation of the line passing through the point $(4, -14, 4)$ and intersecting the line of intersection of the planes : $3x + 2y - z = 5$ and $x - 2y - 2z = -1$ at right angles.

Ans. $\frac{x-4}{3} = \frac{y+14}{10} = \frac{z-4}{4}$

Sol. Direction vector of L.O.I of plane = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -1 \\ 1 & -2 & -2 \end{vmatrix}$

$$= \hat{i}(-4-2) - \hat{j}(-6+1) + \hat{k}(-6-2)$$

$$\vec{P_1} = -6\hat{i} + 5\hat{j} - 8\hat{k}$$

Let $z = 0$

$$3x + 2y = 5$$

$$x - 2y = -1$$

$$x = 1, y = 1$$

$$\text{L.O.I } \frac{x-1}{-6} = \frac{y-1}{5} = \frac{z}{8} = \lambda$$

Let req line intersect at point P $(-6\lambda + 1, 5\lambda + 1, -8\lambda)$

Q $(4, -14, 4)$

$$\vec{PQ} = (-6\lambda - 3, 5\lambda + 15, 8\lambda - 4)$$

$$\vec{PQ} \perp \vec{P_1}$$

$$\vec{PQ} \cdot \vec{P_1} = 0$$

$$-6(-6\lambda - 3) + 5(5\lambda + 15) - 8(-8\lambda - 4) = 0$$

$$36\lambda + 18 + 25\lambda + 75 + 64\lambda + 32 = 0$$

$$125\lambda = -125$$

$$\lambda = -1$$

$$\vec{PQ} = (3, 10, 4)$$

$$\frac{x-4}{3} = \frac{y+14}{10} = \frac{z-4}{4} = \mu$$

Q29. Find the equations to the line which can be drawn from the point $(2, -1, 3)$ perpendicular to the lines $\frac{z-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{z-4}{4} = \frac{y}{5} = \frac{z+3}{3}$ at right angles.

Ans. $\frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2}$

Sol. Req line is perpendicular to given both lines.

direction vector of req. line $= \vec{P} = (\vec{P}_1 \times \vec{P}_2)$

$$\vec{P}_1 \times \vec{P}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 4 & 5 & 3 \end{vmatrix}$$

$$= \hat{i}(a-20) - \hat{j}(6-16) + \hat{k}(10+2)$$

$$= -11\hat{i} + 10\hat{j} - 2\hat{k}$$

$$\text{Equation line} \Rightarrow \frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2} = \mu$$

Q30. Find the value of p so that the lines $\frac{x+1}{-3} = \frac{y-p}{2} = \frac{z+2}{1}$ and $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+2}{2}$ are in the same plane. For this value of p, find the coordinates of their point of intersection and the equation of the plane containing them.

Ans. $p = 3, (2, 1, -3); x + y + z = 0$

Sol. It given line are in same plane

$$\text{then } \begin{vmatrix} -3 & P-7 & 5 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$$-(4+3) - (P-7)(-6-1) + 5(9-2) = 0$$

$$-7 + 7(P-7) + 5(7) = 0$$

$$-1 + P - 7 + 5 = 0$$

$$P = 13$$

$$\text{Equation of plane } \begin{vmatrix} x & y-7 & z+7 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$$x(4+3) - (y-7)(-6-1) + (z+7)(9-2) = 0$$

$$x + y - 7 + 7 + 7 = 0$$

$$x + y + 7 = 0$$

Q31. Let $P = (1, 0, -1)$; $Q = (1, 1, 1)$ and $R = (2, 1, 3)$ are three points.

(a) Find the area of the triangle having P, Q and R as its vertices.

(b) Give the equation of the plane through P, Q and R in the form $ax + by + cz = 1$.

(c) Where does the plane in part (b) intersect the y-axis.

(d) Give parametric equations for the line through R that is perpendicular to the plane in part (b).

Ans. (a) $\frac{3}{2}$; (b) $\frac{2x}{3} + \frac{2y}{3} - \frac{2z}{3} = 1$ (c) $\left(0, \frac{3}{2}, 0\right)$; (d) $x = 2t + 2$; $y = 2t + 1$ and $z = -t + 3$

Sol. $P(1, 0, -1)$ $Q(1, 1, 1)$ $R(2, 1, 3)$

$$(a) P(Q)n = \frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$= \frac{1}{2} \sqrt{4 + 4 + 1}$$

$$= \frac{3}{2}$$

(b) Equation of plane through P, Q, R

$$\therefore \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\vec{a}(1, 0, -1) \cdot \vec{n} = (2, 2, -1)$$

$$2x + 2y - 7 = 2 + 0 + 1$$

$$2x + 2y - 7 = 3 \Rightarrow \frac{2x}{3} + \frac{2y}{3} - \frac{1}{3}7 = 1$$

(c) Plane intersect y-axis at $\left(0, \frac{3}{2}, 0\right)$

(d) Equation of line $\perp$ and passing R

$$\frac{x-2}{2} = \frac{y-1}{2} = \frac{z-3}{1} = t$$

$$x = 2t + 2, y = 2t + 1, z = t + 3$$

Q32. Find the point where the line of intersection of the planes $x - 2y + z = 1$ and $x + 2y - 2z = 5$, intersects the plane $2x + 2y + z + 6 = 0$.

Ans. $(1, -2, -4)$

Sol. Planes are $x - 2y + z = 1$ (i)

$$x + 2y - 2z = 5 \quad \text{....(ii)}$$

$$2x + 2y + z = -6 \quad \text{....(iii)}$$

Add (i) + (ii) + (iii)

$$4x + 2y = 0 \Rightarrow y = -2x \quad \text{....(iv)}$$

From equations (iii) - (i)

$$x + 4y = -7 \quad \text{....(v)}$$

from (iv) and (v) we get

$$x = 1, y = -2$$

Put in (i) we get $z = -4$

So point of intersection is $(1, -2, -4)$

- Q33. Feet of the perpendicular drawn from the point P $(2, 3, -5)$ on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of ΔABC .

Ans. $\frac{x}{2} + \frac{y}{3} + \frac{z}{-5} = 1$, Area = $\frac{19}{2}$ sq. units

Sol. P $(2, 3, -5)$

A $(2, 0, 0)$ B $(0, 3, 0)$ C $(0, 0, -5)$

Equation plane

$$\frac{x}{2} + \frac{y}{3} + \frac{z}{-5} = 1$$

$$\text{Area } \Delta ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & 0 \\ -2 & 0 & -5 \end{vmatrix}$$

$$= \hat{i}(-15) - \hat{j}(10) + \hat{k}(6)$$

$$= -15\hat{i} - 10\hat{j} + 6\hat{k}$$

$$\frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{225 + 100 + 36}$$

$$= \frac{1}{2} \sqrt{361}$$

$$= \frac{19}{2}$$

- Q34. Find the equation of the plane containing the straight line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and perpendicular to the plane $x - y + z + 2 = 0$.

Ans. $2x + 3y + z + 4 = 0$

Sol. Line : $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$

Plane : $x - y + z + 2 = 0$

The vector perpendicular to required plane is $M = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 5 \\ 1 & -1 & 2 \end{bmatrix} = 2\hat{i} + 3\hat{j} + \hat{k}$

Now equation of plane passing through $(1, -2, 0)$ and perpendicular to $2\hat{i} + 3\hat{j} + \hat{k}$

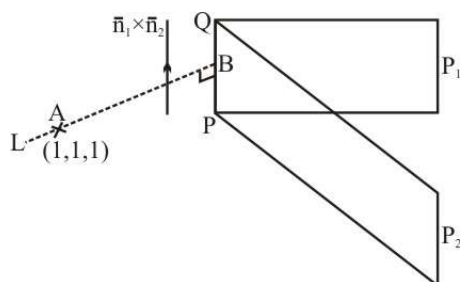
$$(x-1)2 + (y+2)3 + (z-0)1 = 0$$

$$\Rightarrow 2x + 3y + z + 4 = 0$$

- Q35. Let P_1 be a plane passing through two points $A(1, 2, 0)$ and $B(2, -1, 1)$ and perpendicular to $\vec{r} \cdot (\hat{i} + \hat{j} - 3\hat{k}) = 1$. Let L be a line through $(1, 1, 1)$ and intersect line of intersection of planes P_1 and P_2 perpendicularly where equation of plane P_2 is $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5$. If the equation of line L is $\frac{x+1}{-1} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, then find the value of $\left(\frac{2y_1 + z_1}{bc}\right)$.

Ans. 9

Sol.



Equation of plane P

$$\vec{n}_1 = \vec{n} \times \vec{AB} \quad (\hat{i} + \hat{j} - 3\hat{k}) \times (\hat{i} - 3\hat{j} + \hat{k}) = 2\hat{i} + \hat{j} + \hat{k}$$

$$\therefore \text{Equation of plane } P_1 : 2(x-1) + 1(y-2) + 1(z-0) = 0 \Rightarrow P_1 : 2x + y + z = 4$$

$$\text{Also, } P_2 : x + y + z = 5.$$

Line PQ is parallel to vector

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \hat{k} - \hat{j}$$

$$\therefore \text{Equation } PQ : \frac{x+1}{0} = \frac{y-6}{-1} = \frac{z-0}{1} = \lambda$$

Let $B(-1, 6-\lambda, \lambda)$ be any point on line PQ . So, $\vec{AB} = -2\hat{i} + (5-\lambda)\hat{j} + (\lambda-1)\hat{k}$

$$\text{Now, } \vec{AB} \cdot \vec{n}_1 \times \vec{n}_2 = 0 \Rightarrow -5 + \lambda + \lambda - 0 \Rightarrow \lambda = 3$$

$$\therefore B \equiv (-1, 3, 3)$$

$$\text{So, equation of line } L \text{ is } \frac{x+1}{1+1} = \frac{y-1}{1-3} = \frac{z-3}{1-3}$$

$$\text{i.e., } \frac{x+1}{-1} = \frac{y-3}{1} = \frac{z-3}{1}$$

$$\text{Hence } \left(\frac{2y_1}{bc} = \frac{6+3}{1}\right) = 9$$

Q36.

Let L be the line given by $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and let P be the point $(2, -1, 1)$. Also

suppose that E be the plane containing three non collinear points $A = (0, 1, 1)$; $B(1, 2, 2)$ and $C = (1, 0, 1)$

Find

- (a) Distance between the point P and the line L.
- (b) Equation of the plane E.
- (c) Equation the plane F containing the line L and the point P.
- (d) Acute between the plane E and F.
- (e) Volume of the parallelopiped by A, B, C and the point D(- 3, 0, 1).

Ans. (a) $\sqrt{3}$; (b) $x + y - 2z + 1 = 0$; (c) $x - 2y + z = 5$; (d) $\pi/3$; (e) 4

Sol. $\vec{r} = (2, -2, -1) + \lambda (-1, 0, 1)$

$$\vec{a} \quad \vec{P_1}$$

(a) let point on line is Q $(2 - \lambda, -2, -1 + \lambda)$

$$\vec{PQ} = (-\lambda, -1, -2 + \lambda)$$

$$\vec{PQ} \perp \vec{P_1}$$

$$(-\lambda)(-1) + (+)(0) + 1(-2 + \lambda) = 0$$

$$+\lambda - 2 + \lambda = 0$$

$$\lambda = 1$$

$$\vec{PQ} = (-1, -1, -1)$$

$$|\vec{PQ}| = \sqrt{3}$$

(b) $\vec{AB} = (1, 1, 1)$

$$\vec{AC} = (1, -1, 0)$$

$$\text{Equation of plane} = \begin{vmatrix} x-0 & y-1 & z-1 \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = 0$$

$$x(0+1) - (y-1)(0-1) + (z-1)(-1-1) = 0$$

$$x + y - 1 - 2z + 2 = 0$$

$$\text{Equation : } x + y - 2z + 1 = 0$$

(c) Equation of plane containing line 'L' and point 'P'

$$\vec{a}(2, -2, -1) \text{ P}(2, -1, 1)$$

$$\vec{Pa} = (0, -1, -2)$$

$$\begin{vmatrix} x-2 & y+1 & z+1 \\ -1 & 0 & 1 \\ 0 & -1 & -2 \end{vmatrix} = 0$$

$$(x-2)[1] - (y+1)(2) + (z+1)(1) = 0$$

$$x - 2 - 2y - 2 + z + 1 = 0$$

$$F : x - 2y + z = 3$$

$$(d) \vec{n_1} \text{ of } E = (1, 1, -2)$$

$$\vec{n_1} \text{ of } E = (1, -2, 1)$$

$$\cos \theta = \frac{\vec{n_1} \cdot \vec{n_2}}{|\vec{n_1}| |\vec{n_2}|} = \left| \frac{1 - 2 - 2}{\sqrt{6}\sqrt{6}} \right| = \frac{3}{6} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$(e) A(0, 1, 1) B(1, 2, 2) C(1, 0, 1) D(-3, 0, 1)$$

$$\vec{AB} = (1, 1, 1)$$

$$\vec{AC} = (1, -1, 0)$$

$$\vec{AD} = (-3, -1, 0)$$

$$\text{Volume} = [\vec{AB} \quad \vec{AC} \quad \vec{AD}]$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ -3 & -1 & -0 \end{vmatrix} = |(-1 - 3)|$$

$$= 4$$

Q37. Find the foot and hence the length of the perpendicular from the point $(5, 7, 3)$ to the line $\frac{x-15}{3} = \frac{y-29}{8} = \frac{5-z}{5}$. Also find the equation of the plane in which the perpendicular and the given straight line lie.

Ans. $(9, 13, 15)$; 14 ; $9x - 4y - z = 14$

Sol. Let foot on line is $Q(3\lambda + 15, 8\lambda + 2a, 5\lambda + 5)$

$$P(5, 7, 3)$$

$$\vec{PQ} = (3\lambda + 10, 8\lambda + 22, -5\lambda + 2)$$

$$\vec{PQ} \perp \text{line}$$

$$(3\lambda + 10)3 + (8\lambda + 22)8 + (5\lambda + 2)(-5) = 0$$

$$a\lambda + 30 + 64\lambda + 176 + 25\lambda - 10 = 0$$

$$a \& \lambda = 196$$

$$\lambda = -2$$

$$Q(9, 13, 15)$$

$$PQ = (4, 8, 12)$$

$$\overrightarrow{PQ} = \sqrt{16 + 36 + 144}$$

$$= \sqrt{196}$$

Equation plane containing given line and perpendicular (PQ)

$$\begin{vmatrix} x-15 & y-2a & z-5 \\ 3 & 8 & -5 \\ 4 & 6 & -12 \end{vmatrix} = 0$$

$$9x - 4y - z = 14$$

Q38. Let the equation of the plane containing the line $x - y - z - 4 = 0 = x + y + 2z - 4$ and is parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$.

Compute the value of $|A + B + C|$.

Ans. 11

Sol. Direction vector of intersection of given plane

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{vmatrix}$$

$$= \hat{i}(6 - 3) - \hat{j}(4 - 1) + \hat{k}(6 - 3)$$

$$\vec{P} = 3\hat{i} - 3\hat{j} + 3\hat{k}$$

req. plane is

$$(x - y - z - 4) + \lambda (x + y + 2z - 4) = 0$$

$$(1 + \lambda)x + (-1 + \lambda)y + (-1 + 2\lambda)z - 4 - 4\lambda = 0$$

Plane $\pi \perp \vec{P}$

$$3(1 + \lambda) - 3(-1 + \lambda) + 3(-1 + 2\lambda) = 0$$

$$\lambda = \frac{1}{2}$$

$$\frac{x}{2} - \frac{3}{3}y - 2z - 2 = 0$$

$$x - 3y - 4z - 1 = 0$$

$$A = -3, B = -4, C = -1$$

$$|A + B + C| = |-11| = 11$$

Q39. Find the equation of the plane containing the line $\frac{x-1}{2} = \frac{y}{3} = \frac{z}{2}$ and parallel to the line $\frac{x-3}{2} = \frac{y}{5} = \frac{z-2}{4}$. Find also the S.D. between the two lines.

Ans. $x - 2y + 2z - 1 = 0$; 2 units

Sol. $L_1 : \frac{x-1}{2} = \frac{y}{3} = \frac{z}{2}$ $L_2 : \frac{x-3}{2} = \frac{y}{5} = \frac{z-2}{4}$

normal vector of req. plane is $= \vec{P}_1 \times \vec{P}_2$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 2 & 5 & 4 \end{vmatrix} = \hat{i}(2) - \hat{j}(4) + \hat{k}(4)$$

$$= 2\hat{i} - 4\hat{j} + 4\hat{k}$$

Point on L_1 $\vec{a}(1, 0, 0)$ lie in plane

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$2x - 4y + 4z = 2 \Rightarrow x - 2y + 2z = 1$$

Point on L_2 $\vec{b}(3, 0, 2)$

$$\vec{ab} = (2, 0, 2)$$

$$\text{S. D.} = \frac{\vec{n} \cdot \vec{ab}}{|\vec{n}|}$$

$$= \left| \frac{2 + (0)(-2) + 2(2)}{\sqrt{9}} \right|$$

$$= \frac{6}{3} = 2$$

Q40. Consider the plane

$$E : \vec{r} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} + \lambda \vec{r} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \mu \vec{r} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Let F be the plane containing the point A $(-4, 2, 2)$ and parallel to E.

Suppose the point B is on the plane E such that B has a minimum distance from the point A.

If C $(-3, 0, 4)$ lies in the plane F. Find the area of the angle ABC.

Ans. $9/2$

Sol. Normal vector of plane E

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i}(2) - \hat{j} + \hat{k}(-2)$$

$$= 2\hat{i} - \hat{j} - 2\hat{k}$$

$$E : 2x - y - 2z = z - 1 - 2$$

$$2x - y - 2z + 5 = 0$$

B is foot of A in plane E

$$\frac{x+4}{2} = \frac{y-z}{-1} = \frac{z-2}{1} = -\frac{(-8-2-4+5)}{4+1+4}$$

$$\frac{x+4}{2} = \frac{y-2}{-1} = \frac{z-2}{-2} = 1$$

$$B(-2, 1, 0) \quad A(-4, 2, 2) \quad C(-3, 0, 4)$$

$$\Delta ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -2 \\ 1 & -2 & 2 \end{vmatrix} = \hat{i}(-2-4) - \hat{j}(4+2) + \hat{k}(-4+1)$$

$$= -6\hat{i} - 6\hat{j} - 3\hat{k}$$

$$= \frac{1}{2} \sqrt{36 + 36 + 9}$$

$$= \frac{9}{2}$$

- Q41. Let P be the plane passing through (1, 1, 1) and parallel to the lines L_1 and L_2 having direction ratios 1, 0, -1 and -1, 1, 0 respectively. If A, B and C are the points at which P intersects the coordinate axes, find the volume of the tetrahedron whose vertices are A, B, C and the origin.

Ans. $9/2$

Sol. Equation of plane [plane $\mu \perp$ to L_1 & L_2]

$$\begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{vmatrix} = 0$$

$$(x-1)(1) - (y-1)(-1) + (z-1)(1) = 0$$

$$x-1 + y-1 + z-1 = 0$$

$$x + y + z = 3$$

$$A(3, 0, 0), B(0, 3, 0), C(0, 0, 3) \quad D(0, 0, 0)$$

$$\text{Volume} = \frac{1}{6} [\vec{OA} \quad \vec{OB} \quad \vec{OC}]$$

$$= \frac{1}{6} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix}$$

$$= \frac{27}{6} = \frac{9}{2}$$

Q42. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations :

$$3x - y - z = 0 ; \quad -3x + z = 0 ; \quad -3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is

Ans. 7

Sol. $3x - y - z = 0$ (i)

$-3x + z = 0$ (ii)

$-3x + 2y + z = 0$ (iii)

Solving (i) & (ii)

$$y = 0$$

So $3x - z = 0$

$$z = 3x$$

Now $x^2 + y^2 + z^2 \geq 100$

$$x^2 + 9x^2 \leq 100$$

$$-\sqrt{10} \leq x \leq \sqrt{10}$$

Integral values of 'x' are -3, -2, -1, 0, 1, 2, 3

So '7' points are there

Q43. If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines

$$\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{4} \text{ and } \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \text{ is } \sqrt{6}, \text{ then } |d| \text{ is}$$

Ans. 6

Sol. Plane containing the line

Direction ratio's of normal to the plane :

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = -\hat{i} + 2\hat{j} - \hat{k}$$

Hence equation of plane $1(x - 1) - 2(y - 2) + 1(z - 3) = 0$

i.e. $x - 2y + z = 0$

As given plane must be parallel $\Rightarrow A = 1$

& distance between the planes

$$\left| \frac{d - 0}{\sqrt{1^2 + 2^2 + 1^2}} \right| = \sqrt{6}$$

$$|d| = 6$$

Q44. The plane denoted by $\Pi_1: 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2: 5x + 3y + 10z = 25$. If the plane in its new position be denoted by Π , and the distance of this plane from the origin is $\sqrt{k}$ where $k \in \mathbb{N}$, then find k .

Ans. 212

Sol. Req plane $\pi + \lambda \pi_2 = 0$

$$\pi: (4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + 81 - 25\lambda = 0$$

$$\pi \perp \pi_1$$

$$4(4 + 5\lambda) + 7(7 + 3\lambda) + 4(4 + 10\lambda) = 0$$

$$16 + 20\lambda + 49 + 21\lambda + 16 + 40\lambda = 0$$

$$81 + 81\lambda = 0$$

$$\lambda = -1$$

$$\pi: -x + 4y - 6z + 106 = 0$$

distance from origin

$$d = \frac{106}{\sqrt{1 + 16 + 36}} = \frac{106}{\sqrt{53}} = \sqrt{212}$$

Q45. Find the equation of the plane containing the line $2x - y + z - 3 = 0$, $3x + y + z = 5$ and at a distance of $1/\sqrt{6}$ from the point $(2, 1, -1)$.

Ans. $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

Sol. Equation of plane containing the line,

$$2x - y + z - 3 = 0 \text{ and } 3x + y + z = 5 \text{ is}$$

$$(2x - y + z - 3) + \lambda(3x + y + z - 5) = 0$$

$$\Rightarrow (2 + 3\lambda)x + (\lambda - 1)y + (\lambda + 1)z - 3 - 5\lambda = 0$$

Since distance of plane from $(2, 1, -1)$ to above plane is $1/\sqrt{6}$

$$\therefore \left| \frac{6\lambda + 4 + \lambda - 1 - 3 - 5\lambda}{\sqrt{(3\lambda + 2)^2 + (\lambda - 1)^2 + (\lambda + 1)^2}} \right| = \frac{1}{\sqrt{6}}$$

$$\Rightarrow 6(\lambda - 1)^2 = 11\lambda^2 + 12\lambda + 6$$

$$\Rightarrow \lambda = 0, -\frac{24}{5}$$

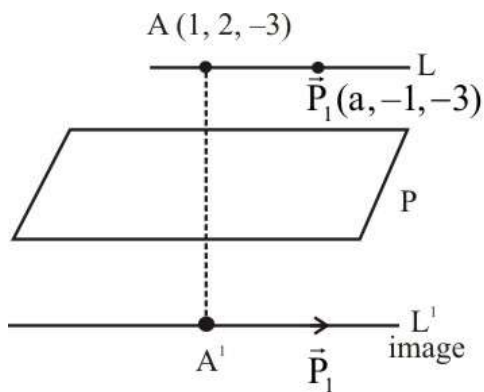
$\therefore$ Equation of planes are,

$$2x - y + z - 3 = 0 \text{ and } 62x + 29y + 19z - 105 = 0$$

Q46. Find the equation of the line which is reflection of the line $\frac{x-1}{9} = \frac{y-2}{-1} = \frac{z+3}{-3}$ in the plane $3x - 3y + 10z = 26$.

Ans. $\frac{x-4}{9} = \frac{y+1}{-1} = \frac{z-7}{-3}$

Sol.



$$L : \frac{x-1}{9} = \frac{y-2}{-1} = \frac{z+3}{-3}$$

$$P : 3x - 3y + 10z = 26$$

$$L \parallel P$$

A^1 is image of A in the given plane :

$$\begin{aligned} \frac{x-1}{3} = \frac{y-2}{-3} = \frac{z+3}{10} &= -2 \left(\frac{3-6-30-26}{9+9+100} \right) \\ &= -2 \left(\frac{-59}{118} \right) \end{aligned}$$

$$\frac{x-1}{3} = \frac{y-2}{-3} = \frac{z+3}{10} = +1$$

$$A^1 (4, -1, 7)$$

$$L^1 : \frac{x-4}{9} = \frac{y+1}{-1} = \frac{z-7}{-3} = \lambda$$

Q47. Find the equations to the line of greatest slope through the point $(7, 2, -1)$ in the plane $x - 2y + 3z = 0$ assuming that the axes are so placed that the plane $2x + 3y - 4z = 0$ is horizontal.

Ans. $\frac{z-7}{22} = \frac{y-2}{5} = \frac{z+1}{-1}$

Sol. $\pi_G : x - 2y + 37 = 0$

$\pi_H : 2x - 3y - 47 = 0$

$n_g (1, -2, 3), \quad n_h (2, 3, -4)$

Direction vector of Req. line $\vec{p} = \vec{n}_g \times (\vec{n}_g \times \vec{n}_h)$

$$= \vec{n}_g (\vec{n}_g \cdot \vec{n}_h) - \vec{n}_h (\vec{n}_g \cdot \vec{n}_g)$$

$$= (1, -2, 3) (-16) - (2, 3, -4) (14)$$

$$= (-2) (22, 5, -4)$$

$$\frac{x-7}{22} = \frac{y-2}{5} = \frac{z+4}{-1} = \lambda$$

Paragraph

Let A (1, 2, 3), B (0, 0, 1), C(-1, 1, 1) are the vertices of a ΔABC

\n

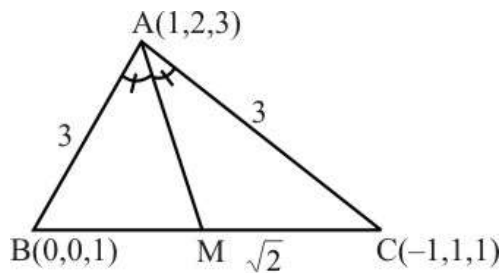
Q48. The equation of internal angle bisector through A to side BC is

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$ (B) $\vec{r} = -\hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$

(C) $\vec{r} = -\hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$ (D) $\vec{r} = -\hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

Ans. D

Sol.



Here ABC is isosceles with $AB = AC$,

So internal bisector of A is perpendicular to BC

As $\Delta AMB \cong \Delta AMC$ (R.H.S. rule)

$\Rightarrow$ M is midpoint of BC

So, $M \equiv \left(\frac{-1}{2}, \frac{1}{2}, 1 \right)$

$\therefore$ Equation of internal bisector through A to side BC is

$$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \mu \left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k} \right) \text{ OR } \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu (3\hat{i} + 3\hat{j} + 4\hat{k})$$

Alternative for (i) Equation of BC is

$$\vec{r} = \hat{k} + \lambda(\hat{i} - \hat{j})$$

Let p.v. of M on BC be $\vec{r}$

Now, $\overrightarrow{AM} = \text{p. v. of M} - \text{p.v of A} = (\lambda\hat{i} - \lambda\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 2\hat{k}$

Since $\overrightarrow{AM} \cdot (\hat{i} - \hat{j}) = 0 \Rightarrow \lambda = -1/2$

$\therefore$ p.v. of point M = $\frac{-1}{2}\hat{i} + \frac{1}{2}\hat{j} + \hat{k}$

$\Rightarrow$ Equation of internal bisector through A to side BC is

$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \mu \left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k} \right)$ OR $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu (3\hat{i} + 3\hat{j} + 4\hat{k})$

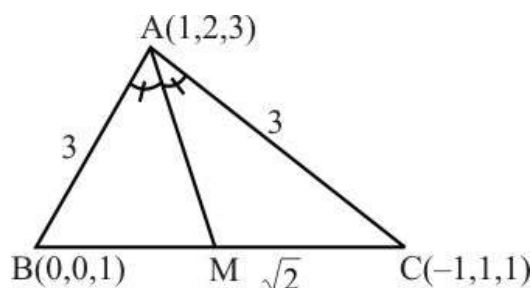
Q49. The equation of median through C to side AB is

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$ (B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$

(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$ (D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{j})$

Ans. B

Sol.



Clearly mid point L of AB is $\left(\frac{1}{2}, 2 \right)$

$\Rightarrow$ Equation of median through C to AB is

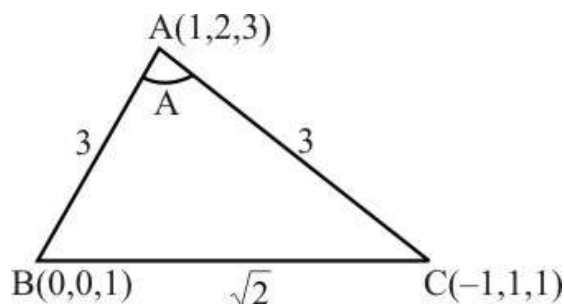
$\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p \left(\frac{3}{2}\hat{i} + \hat{k} \right)$ OR $\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p(3\hat{i} + 2\hat{k})$

Q50. The area (ΔABC) is equal to

(A) $\frac{9}{2}$ (B) $\frac{\sqrt{17}}{2}$ (C) $\frac{17}{2}$ (D) $\frac{7}{2}$

Ans. B

Sol.



$$\text{We have } \cos A = \frac{3^2 + 3^2 - (\sqrt{2})^2}{2(3)(3)}$$

$$\therefore \cos A = \frac{16}{18} = \frac{8}{9}$$

$$\text{Now area } (\Delta ABC) = (3)(3) \sin A = \frac{9}{2} \sqrt{1 - \frac{64}{81}} = \frac{9}{2} \times \frac{\sqrt{17}}{9} = \frac{\sqrt{17}}{2} \text{ sq. unit}$$

Paragraph

Consider a plane

\n $x + y - z = 1$ and the point $A(1, 2, -3)$

\n A line L has the equation

\n $x = 1 + 3r, \quad y = 2 - r, \quad z = 3 + 4r$

\n

Q51. The co-ordinate of a point B of line L, such that AB is parallel to the plane, is

(A) $10, -1, 15$ (B) $-5, 4, -5$ (C) $4, 1, 7$ (D) $-8, 5, -9$

Ans. D

Sol. Line $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = r$; $A(1, 2, -3)$

Any point say $B \equiv 3r + 1, 2 - r, 3 + 4r$ (on the line L)

$$\overline{AB} = 3r, -r, 4 + 6$$

Hence $\overline{AB}$ is parallel to $x + y - z = 1$

$$\text{hence, } 3r - r - 4r - 6 = 0$$

$$2r = -6 \quad \Rightarrow \quad r = -3$$

hence B is $-8, 5, -9$

Q52. Equation of the plane containing the line L and the point A has the equation

(A) $x - 3y + 5 = 0$ (B) $x + 3y - 7 = 0$ (C) $3x - y - 1 = 0$ (D) $3x + y - 5 = 0$

Ans. B

Sol. Equation of plane containing the line L is

$$A(x - 1) + B(y - 2) + C(z - 3) = 0, \text{ where } 3A - B + 4C = 0 \quad \dots\dots(1)$$

\therefore (1) also contains the point $A(1, 2, -3)$

$$\text{hence } C = 0; 3A = B$$

$$\text{equation of plane } x - 1 + 3(y - 2) = 0$$

$$x + 3y - 7 = 0$$

Paragraph

Consider the lines $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}; \quad L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

\n

Q53. The shortest distance between L_1 and L_2 is

- (A) 0 (B) $\frac{17}{\sqrt{3}}$ (C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

Ans. D

Sol. Shortest distance

$$= (3\mathbf{i} - 4\mathbf{k}) \cdot \frac{(-\mathbf{i} - 7\mathbf{j} + 5\mathbf{k})}{5\sqrt{3}} = \frac{17}{5\sqrt{3}}$$

Q54. The distance of the point (1, 1, 1) from the plane passing through the point (-1, -2, -1) and whose normal is perpendicular to both the lines L_1 and L_2 is

- (A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$ (C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Ans. C

Sol. Equation of plane $-(x + 1) - 7(y + 2) + 5(z + 1) = 0$

$$x + 7y - 5z + 10 = 0$$

$$\text{distance from } (1, 1, 1) = \frac{1 + 7 - 5 + 10}{5\sqrt{3}} = \frac{13}{5\sqrt{3}}$$

Q55. The unit vector perpendicular to both L_1 and L_2 is

- (A) $\frac{-\hat{i} + 7\hat{j} + 7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$ (C) $\frac{-\hat{i} + 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i} - 7\hat{j} - \hat{k}}{\sqrt{99}}$

Ans. B

Sol. $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$

$$L_2 : \frac{x-2}{3} = \frac{y+2}{2} = \frac{z-3}{3}$$

a vector perpendicular to L_1 & L_2 will be

$$\text{Hence unit vector} = \frac{-\mathbf{i} - 7\mathbf{j} + 5\mathbf{k}}{5\sqrt{3}}$$

Q56. A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point P(1, 0) can be

- (A) $6\hat{i} + 8\hat{j}$ (B) $-6\hat{i} + 8\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $-6\hat{i} - 8\hat{j}$

Ans. A

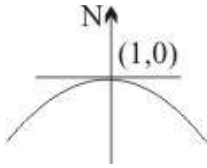
Sol. differentiate the curve

$$6x + 8(xy_1 + y) + 4yy_1 = 0$$

$$m_T \text{ at } (1, 0) \text{ is } 6 + 8(y_1(1)) = 0$$

$$y_1(0) = -\frac{3}{4}$$

$$m_N = \frac{4}{3} \Rightarrow \text{vector along normal } 3\hat{i} + 4\hat{j}$$



$$\text{unit vector} = \pm \frac{(3\hat{i} + 4\hat{j})}{5} \text{ along the normal}$$

$$\text{again normal vector of magnitude } 10 = \pm (6\hat{i} + 8\hat{j})$$

Q57. Consider a plane P passing through $A(\lambda, 3, \mu)$, $B(-1, 3, 2)$ and $C(7, 5, 10)$ and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5

(B) Equation of straight line L_1 is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$

Ans. B, C, D

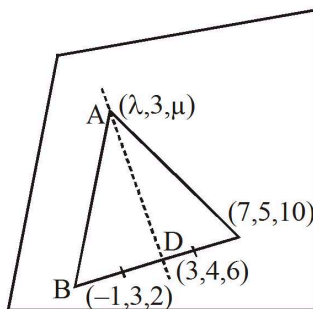
Sol. (A) Dir's of Straight L which is passing through A and bisecting BC are $\lambda - 3, -1, \mu - 6$.

Straight line L is equally inclined with co-ordinate axes

$$\therefore \frac{\lambda - 3}{1} = \frac{-1}{1} = \frac{\mu - 6}{1} \Rightarrow \lambda = 2 \text{ and } \mu = 2$$

$$\lambda + \mu = 7 \text{ Ans.}$$

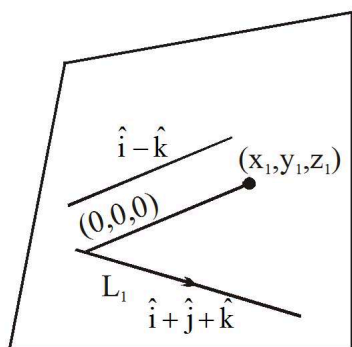
(B) Dir's of line L are 1, 1, 1



$\therefore$ Option 'c' is correct

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1} \text{ it satisfied } (0, 0, 0)$$

C) Normal vector of plane P is



$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & 3 \\ -5 & -2 & -5 \end{vmatrix} = 6\hat{i} - \hat{j} \times 0 + \hat{k}(-6) = 6(\hat{i} - \hat{k}) = .$$

Plane perpendicular to plane P and containing line L_1 is

$$= \begin{vmatrix} x & y & z \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow x(0+1) - y(1+1) + z(1) = 0$$

$$\Rightarrow x - 2y + z = 0. \text{ Ans.}$$

Q58. A line L passing through the point P(1, 4, 3), is perpendicular to both the lines

$$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{2} \text{ and } \frac{x+2}{3} = \frac{y+4}{2} = \frac{z+1}{-2}.$$

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be

(A) 16 (B) 15 (C) 2 (D) 1

Ans. B, D

Sol. Equation of the line passing through P(1, 4, 3) is

$$\frac{x-1}{a} = \frac{y-4}{b} = \frac{z-3}{c} \quad \dots(1)$$

where $a\hat{i} + b\hat{j} + c\hat{k}$ is along the common perpendicular to both the given lines.

Since (1) is perpendicular to $\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{2}$ and $\frac{x+2}{3} = \frac{y+4}{2} = \frac{z+1}{-2}$

$$\text{Hence } 2a + b + 4c = 0$$

$$\text{and } 3a + 2b - 2c = 0$$

$$\therefore \frac{a}{-2-8} = \frac{b}{12+4} = \frac{c}{4-3} \Rightarrow \frac{a}{-10} = \frac{b}{16} = \frac{c}{1}$$

$$\text{Hence the equation of the lines is } \frac{x-1}{-10} = \frac{y-4}{16} = \frac{z-3}{1} \quad \dots(2) \text{ Ans.}$$

Now any point Q on (2) can be taken as $(1 - 10\lambda, 16\lambda + 4, \lambda + 3)$

$$\therefore \text{Distance of Q from P (1, 4, 3)} = (10\lambda)^2 + (16\lambda)^2 + \lambda^2 = 357$$

$$\Rightarrow (100 + 256 + 1)\lambda^2 = 357$$

$$\Rightarrow \lambda = 1 \text{ or } -1$$

∴ Q is (-9, 20, 4) or (11, -12, 2)

Hence $a_1 + a_2 + a_3 = 15$ or 1 Ans.]

Q59. If $\vec{a}, \vec{b}, \vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ & $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?

- (A) $\vec{a}, \vec{b}, \vec{c}$ are the orthogonal in pairs (B) $\left[\vec{a} \vec{b} \vec{c} \right] = |\vec{b}|$ (C) $\left[\vec{a} \vec{b} \vec{c} \right] = |\vec{c}|$ (D) $|\vec{b}| = |\vec{c}|$

Ans. C

Sol. Clearly $\vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = 0$ Also $\vec{b} \times \vec{c} = \vec{a} \Rightarrow$ A

$$\vec{a} \times \vec{b} = \vec{c}$$

$$\text{dot with } \vec{b} \Rightarrow \vec{b} \cdot \vec{c} = 0 ; \quad ||| \text{ly } \vec{b} \times \vec{c} = \vec{a}$$

$$\text{dot with } \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0 ; \quad \text{dot with } \vec{c} \Rightarrow \vec{a} \cdot \vec{c} = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0 \quad |\vec{a}| |\vec{b}| = |\vec{c}| |\vec{b}| |\vec{c}| = |\vec{a}| \Rightarrow \frac{|\vec{b}|}{|\vec{c}|} = \frac{|\vec{c}|}{|\vec{a}|} \Rightarrow |\vec{a}| = |\vec{c}| \text{ \& } |\vec{b}| = 1$$

$$\Rightarrow \vec{a} \times \vec{b} \cdot \vec{c} = |\vec{a}| |\vec{b}| |\vec{c}| = |\vec{a}|^2 = |\vec{c}|^2$$

(children will assume $\vec{a} = \hat{i}; \vec{b} = \hat{j}$ and $\vec{c} = \hat{k}$ but in this case all the four will be correct which will be wrong)

Q60. Given the following information about the non zero vectors $\vec{A}, \vec{B}$ and $\vec{C}$

$$(i) (\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$$

$$(ii) \vec{B} \cdot \vec{B} = 4$$

$$(iii) \vec{A} \cdot \vec{A} = 8.$$

$$(iv) \vec{B} \cdot \vec{C} = 6$$

Which of one of the following holds good ?

$$(A) \vec{A} \times \vec{B} = \vec{0} \quad (B) \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{0} \quad (C) \vec{A} \cdot \vec{A} = 8 \quad (D) \vec{A} \cdot \vec{C} = -9$$

Ans. bonus

Sol. Given $\cos \frac{|\vec{A}|}{|\vec{B}|} \cos \theta = - \frac{|\vec{B}|}{6} ; = 2$ (given)

$$\vec{B} \cdot \vec{C} = |\vec{B}| |\vec{C}| = \cos \phi = 6$$

$$\text{and } (\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$$

$$(\vec{A} \cdot \vec{A}) \vec{B} - (\vec{B} \cdot \vec{A}) \vec{A} = \vec{0}$$

$$(\vec{A} \cdot \vec{A}) \vec{B} = -6 \vec{A} \quad \dots\dots\dots(1)$$

$\therefore \vec{A}$ and $\vec{B}$ are collinear and between $\vec{A}$ and $\vec{B}$ is π is correct $\Rightarrow \vec{A} \times \vec{B} = 0 \Rightarrow A$ is correct

$$\Rightarrow \vec{A} \cdot (\vec{A} \times \vec{B}) \cdot \vec{C} = (\vec{A} \times \vec{B}) \cdot \vec{C} = 0 \Rightarrow B \text{ is correct}$$

$$\text{also } \vec{A} \cdot \vec{B} = -6 \text{ and } |\vec{B}| = 2$$

$$\therefore |\vec{A}| |\vec{B}| \cos \pi = -6$$

$$|\vec{A}| \cdot (2) = 6 \Rightarrow |\vec{A}| = 3 \Rightarrow \vec{A} \cdot \vec{A} = 9 \quad C \text{ is not correct}$$

$$\text{again } \vec{A} \cdot \vec{C} = ?$$

dot with the equation (1)

$$9(\vec{B} \cdot \vec{C}) = -\vec{A} \cdot \vec{C}$$

$$9(6) = -6(\vec{A} \cdot \vec{A}) \Rightarrow \vec{A} \cdot \vec{C} = -9 \Rightarrow (D) \text{ is not correct}$$

Q61. If $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

$$(A) (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = 0 \quad (B) (\vec{A} \times \vec{C}) \cdot (\vec{B} \times \vec{D}) \neq 0 \quad (C) (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{0}$$

$$(D) (\vec{A} \times \vec{C}) \cdot (\vec{B} \times \vec{D}) \neq 0$$

Ans. B, C

Sol. Obviously (B) and (C)]

Q62. If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, then the vectors $\vec{a} - \vec{d}$ and $\vec{b} - \vec{c}$ are:

$$(A) \text{ collinear} \quad (B) \text{ linearly independent} \quad (C) \text{ perpendicular} \quad (D) \text{ parallel}$$

Ans. A, D

$$\text{Sol. } \vec{a} \times \vec{b} = \vec{c} \times \vec{d} \dots\dots\dots(1)$$

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{d} \dots\dots\dots(2)$$

$$(1) - (2)$$

$$\vec{a} \times (\vec{b} - \vec{c}) = -(\vec{b} - \vec{c}) \times \vec{d}$$

$$(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = 0$$

$$(\vec{a} - \vec{d}) \parallel (\vec{b} - \vec{c})$$

Q63. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are linearly independent set of vectors and $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = \vec{0} = 0$, then

- (A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$ (C) $K_1 + K_4 = K_2 + K_3 = 0$
 (D) none of these

Ans. A, B, C

Sol. If a, b, c, α are linearly independent vector then $k_1 = k_2 = k_3 = k_4 = 0$

Q64. If $A(\bar{a})$; $B(\bar{b})$; $C(\bar{c})$ and $D(\bar{d})$ are four points such that

$$\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}; \quad \bar{b} = 2\hat{i} - 8\hat{j}; \quad \bar{c} = \hat{i} - 3\hat{j} + 4\hat{k}; \quad \bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$$

d is the shortest distance between the lines AB and CD, then which of the following is True?

- (A) $d = 0$, hence AB and CD intersect (B) $d = \frac{[\vec{AB} \vec{CD} \vec{BD}]}{|\vec{AB} \times \vec{CD}|}$
 (C) AB and CD are skew lines and $d = \frac{23}{13}$ (D) $d = \frac{[\vec{AB} \vec{CD} \vec{AC}]}{|\vec{AB} \times \vec{CD}|}$

Ans. B, C, D

Sol. $\vec{AB} = \bar{b} - \bar{a} = 4\hat{i} - 12\hat{j} - 3\hat{k}$

$$\vec{CD} = \bar{d} - \bar{c} = 3\hat{i} + 4\hat{j} - 12\hat{k}$$

$$\vec{AC} = \bar{c} - \bar{a} = 3\hat{i} - 7\hat{j} + 2\hat{k}$$

$$\vec{BD} = \bar{d} - \bar{b} = 2\hat{i} + 9\hat{j} - 7\hat{k}$$

By definition $d = \frac{(\vec{AB} \times \vec{CD}) \cdot \vec{AC}}{|\vec{AB} \times \vec{CD}|} \dots\dots(i)$

$$= \frac{(\vec{AB} \times \vec{CD})}{|\vec{AB} \times \vec{CD}|} \dots\dots(ii)$$

$$\vec{AB} \times \vec{CD} = 13(12\hat{i} + 3\hat{j} + 4\hat{k}) \quad \therefore |\vec{AB} \times \vec{CD}| = 169$$

$$\therefore d = \frac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (3\hat{i} - 7\hat{j} + 2\hat{k}) = \frac{23}{13} \quad \text{using (i)}$$

$$\text{also } d = \frac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (2\hat{i} - 9\hat{j} + 7\hat{k}) = \frac{23}{13} \quad \text{using (ii)}$$

Q65. Given the equations of the line $3x - y + z + 1 = 0$, $5x + y + 3z = 0$. Then which of the following is correct ?

- (A) symmetrical form of the equations of line is $= \frac{x}{2} = \frac{y - \frac{1}{8}}{-1} = \frac{z + \frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and perpendicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through (2, 1, 4) and perpendicular to the given lines is $x + y - 2z + 5 = 0$

Ans. B, D

Sol. Let a, b, c be the d.r's. of the given line. Then

$$\text{we have } 3a - b + c = 0$$

$$5a + b + 3c = 0$$

$$\text{Solving we get } \frac{a}{1} = \frac{b}{1} = \frac{c}{-2}$$

Again, suppose the given line intersect the plane $z = 0$ at $(x_1, y_1, 0)$ then $3x_1 - y_1 + 1 = 0$ and $5x_1 + y_1 = 0$

$$\text{Solving we get } x_1 = -\frac{1}{8}, y_1 = \frac{5}{8}$$

Hence the symmetrical form of the line

$$\text{Equation of plane through (2, 1, 4) is } \frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$$

$$A(x - 2) + B(y - 1) + c(z - 4) = 0$$

when $A = 1, b = 1$ and $c = -2$

$$\Rightarrow x - 2 + y - 1 - 2(z - 4) = 0 \quad \Rightarrow \quad x + y - 2z + 5 = 0$$

Q66. Given three vectors

$$\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}; \quad \vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}; \quad \vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$$

Which of the following hold good for the vectors $\vec{U}, \vec{V}$ and $\vec{W}$?

$$(A) \vec{U}, \vec{V} \text{ are } \vec{W} \text{ linearly dependent} \quad (B) (\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$$

$$(C) \vec{U}, \vec{V} \text{ and } \vec{W} \text{ form a triplet of mutually perpendicular vectors} \quad (D) \vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$$

Ans. D, C, B

Sol.

$$\text{If may be observed that } [\vec{U} \vec{V} \vec{W}] = \begin{vmatrix} 2 & 3 & -6 \\ 6 & 2 & 3 \\ 3 & -6 & -2 \end{vmatrix} = 343 \neq 0$$

$$\Rightarrow \vec{U}, \vec{V}, \vec{W} \text{ are non coplanar hence linearly independent}$$

$$\text{Further } \vec{U} \times \vec{V} = \vec{W} \text{ and } \vec{V} \times \vec{W} = \vec{U}$$

$\Rightarrow$ They form a right handed triplet of mutually perpendicular vectors and course !

Q67. Consider the following four pairs of lines in column-I and match them with one or more entries in column-II.

Column-I	Column-II
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = \lambda (2, 2, -10)$	(P) non coplanar lines
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z-2}{3}$	(Q) lines lie in a unique plane
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) infinite planes containing both the lines
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines do not intersect
(A) (A) R (B) Q (C) QS (D) PS	(B) (A) Q (B) R (C) QS (D) PS
(C) (A) R (B) Q (C) PS (D) QS	(D) (A) R (B) QS (C) Q (D) PS

Ans. A

Sol. (A) $L_1 : \frac{x-1}{1} = \frac{y-0}{1} = \frac{z-2}{-5}; \vec{V}_1 = \hat{i} + \hat{j} - 5\hat{k}$
 $L_2 : \frac{x-2}{2} = \frac{y-1}{2} = \frac{z+3}{-10}; \vec{V}_2 = 2(\hat{i} + \hat{j} - 5\hat{k})$

Hence lines are parallel and both contains the points (1, 0, 2) and (2, 1, -3) $\Rightarrow$ coincident line both L_1 and L_2 may lie in an infinite number of planes hence $\Rightarrow$ (R)

(B) $\vec{V}_1 = 2\hat{i} + 2\hat{j} - \hat{k}$
 $\vec{V}_2 = \hat{i} - 2\hat{j} + 3\hat{k}$ $\Rightarrow$ lines not parallel

Also both intersect at (3, 5, 1)

Hence lines are intersecting hence they lie on a unique plane $\Rightarrow$ (P)

(C) $L_1 : \frac{x-0}{-6} = \frac{y-1}{9} = \frac{z-0}{1} = s$

$L_2 : \frac{x-1}{2} = \frac{y-4}{-3} = \frac{z-0}{1} = s$ $\left. \begin{array}{l} L_1 \text{ is parallel to } -6\hat{i} + 9\hat{j} - 3\hat{k} \\ L_2 \text{ is parallel to } 2\hat{i} - 3\hat{j} + \hat{k} \end{array} \right\}$ lines parallel but not coincident

as (0, 1, 0) does not line on L_2 , not intersecting

Hence L_1, L_2 , lies in a unique planes $\Rightarrow$ (Q), (S)

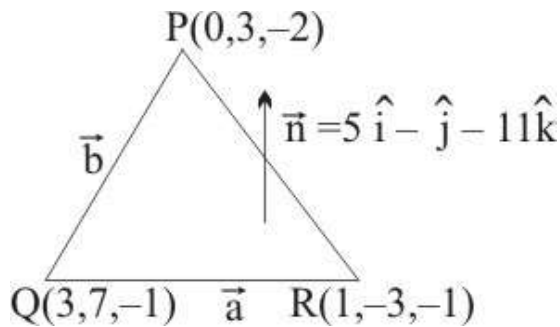
(D) Lines are skew can be verified $\Rightarrow$ (P), (S)]

Q68. $P(0, 3, -2)$; $Q(3, 7, -1)$ and $R(1, -3, -1)$ are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$.

Column-I	Column-II
(A) perpendicular distance of P from L_2	(P) $7\sqrt{3}$
(B) shortest distance between L_1 and L_2	(Q) 2
(C) area of the triangle PQR	(R) 6
(D) distance from $(0, 0, 0)$ to the plane PQR	(S) $\frac{19}{\sqrt{147}}$
(A) (A) Q (B) R (C) P (D) S (B) (A) R (B) Q (C) P (D) S	
(C) (A) R (B) Q (C) S (D) P (D) (A) R (B) P (C) Q (D) S	

Ans. 5efb06ad9fff15766e32d51d

Sol.



$$L_1 : \frac{x}{3} = \frac{y-3}{4} = \frac{z+2}{1} \quad \dots\dots\dots(1)$$

(passing through P and Q)

$$L_2 : \frac{x}{-11} - 11 = \frac{y-3}{0} = \frac{z+1}{1} \quad \dots\dots\dots(2)$$

(passing through R and parallel to $\vec{v} = \hat{i} + \hat{k}$)

(A) $\frac{\text{distance of } P(0, 3, -2) \text{ from } L_2}{\overline{PN} = (t+1)\hat{i} - 6\hat{j} + 2(t-1)\hat{k}}$

now $\overline{PN} \cdot \vec{V} = 0 \Rightarrow [(t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}] \cdot (\hat{i} + \hat{k}) = 0$

$$(t+1) + (t+1) = 0 \quad \Rightarrow \quad t = -1$$

hence $\overline{PN} = -6\hat{j}$

$$|PN| = |-6\hat{j}| = 6 \quad \Rightarrow \quad \text{(R)}$$

Distance between L_1 and L_2

Equation of plane containing L_1 and parallel to L_2

$$Ax + B(y-3) + C(z+2) = 0$$

where $3A + 4B + C = 0$

$$\text{and } A + 0B + C = 0 \Rightarrow A + C = 0$$

$$C = 1, \quad A = -\lambda, \quad B = +\lambda/2$$

$\therefore$ equation of plane

$$-\lambda x + \frac{\lambda}{2}(y - 3) + \lambda(z + 2) = 0$$

$$2x - y + 3 - 2z - 4 = 0$$

$$2x - y - 2z = 1 \quad \dots\dots\dots(1)$$

now distance of the point $(1, -3, -1)$ lying on the line L_2 from the plane (1)

$$d = \left| \frac{2 + 3 + 2 - 1}{3} \right| = 2 \Rightarrow \quad (Q)$$

Area of ΔPQR

$$\overrightarrow{QR} = \vec{a} = 2\hat{i} + 10\hat{j} + 0\hat{k}$$

$$\overrightarrow{OP} = \vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$$

$$\vec{a} \times \vec{b} = 2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 5 & 0 \\ 3 & 4 & 1 \end{vmatrix} = 2[\hat{i}(5) - \hat{j}(1) + \hat{k}(4 - 15)] = 2[5\hat{i} - \hat{j} - 11\hat{k}]$$

$$\frac{|\vec{a} \times \vec{b}|}{2} = \sqrt{25 + 1 + 121} = \sqrt{147} = \sqrt{3 \cdot 49} = 7\sqrt{3} \Rightarrow \quad (P)$$

Distance of $(0, 0, 0)$ from PQR

equation of plane PQR is

$$(\vec{r} - \vec{p}) \cdot \vec{n}$$

$$= [x\hat{i} + (y - 3)\hat{j} + (z + 2)\hat{k}] \cdot [5\hat{i} - \hat{j} - 11\hat{k}]$$

$$= 5x - (y - 3) - 11(z + 2) = 0$$

$$= 5x - y - 11z - 19 = 0$$

distance from $(0, 0, 0)$ of the plane

$$d = \left| \frac{19}{\sqrt{25 + 1 + 121}} \right| = \frac{19}{\sqrt{147}} \Rightarrow \quad (S)$$

Q69.

Column-I

Column-II

(A) Let O be an interior point of ΔABC such that $\overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \vec{0}$, (P) 0

then the ratio of the area of ΔABC to the area of ΔAOC , is
with O is the origin

(B) Let ABC be a triangle whose centroid is G, orthocentre is H and

(Q) 1

circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that no three of O, A, B, C and D are

(R) 2

collinear satisfying the relation $\vec{AD} + \vec{BD} + \vec{CD} = 3\vec{HD} = \lambda\vec{HD}$

(S) 3

then the value of the scalar ' λ ' is

(C) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are non zero vectors such that no three of them are in the same plane and no two are orthogonal then the value of the scalar

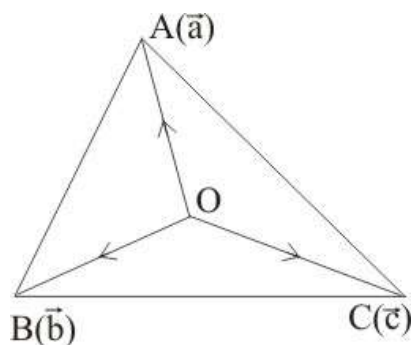
$$\frac{(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d})}{(\vec{a} \times \vec{b}) \cdot (\vec{d} \times \vec{c})} \text{ is}$$

(A) (A) Q (B) R (C) S (B) (A) S (B) Q (C) R (C) (A) S (B) R (C) Q

(D) (A) R (B) Q (C) S

Ans. 5efb0b799fff15766e32d5d4

Sol.



$$(A) \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{\frac{1}{2} |\vec{a} \times \vec{b} + \vec{c} + \vec{c} \times \vec{a}|}{\frac{1}{2} |\vec{a} \times \vec{c}|}$$

$$\text{now } \vec{a} + 2\vec{b} + 3\vec{c} = 0$$

$$\text{cross with } \vec{b}, \vec{a} \times \vec{b} + 3\vec{c} \times \vec{b} = 0 \Rightarrow \vec{a} \times \vec{b} = 3(\vec{b} \times \vec{c})$$

$$\text{cross with } \vec{a}, 2\vec{a} \times \vec{b} + 3\vec{a} \times \vec{c} = 0 \Rightarrow \vec{a} \times \vec{b} = \frac{3}{c}(\vec{c} \times \vec{a})$$

$$\therefore \vec{a} \times \vec{b} = \frac{3}{2}(\vec{c} \times \vec{a}) = 3(\vec{b} \times \vec{c})$$

$$\text{Let } (\vec{c} \times \vec{a}) = \vec{p}$$

$$\vec{a} \times \vec{b} = \frac{3\vec{p}}{2}$$

$$(B) \text{ LHS} = \vec{d} - \vec{a} + \vec{d} - \vec{b} + \vec{h} - \vec{c} + 3(\vec{g} - \vec{h})$$

$$= 2\vec{d} - (\vec{a} + \vec{b} + \vec{c}) + 3\vec{g} - 2\vec{h}$$

$$= 2\vec{d} - 2\vec{h} = 2(\vec{d} - \vec{h}) = 2\vec{HD} \Rightarrow \lambda = 2 \Rightarrow (R)$$

$$(C) (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{d} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{d} \end{vmatrix}$$

|||ly compute others which given (1) $\Rightarrow$ (Q)

Q70. Consider the following linear equations

$$ax + by + cz = 0, \quad bx + cy + az = 0, \quad cx + ay + bz = 0$$

Match the conditions/ expressions in **Column I** with statements in **Column II**.

Column I

Column II

- | | |
|---|---|
| (A) $a + b + c \neq 0$ and
$a^2 + b^2 + c^2 = ab + bc + ca$ | (P) the equation represent planes
meeting only at a single point. |
| (B) $a + b + c = 0$ and
$a^2 + b^2 + c^2 \neq ab + bc + ca$ | (Q) the equation represent the line
$x = y = z$ |
| (C) $a + b + c \neq 0$ and
$a^2 + b^2 + c^2 \neq ab + bc + ca$ | (R) the equation represent identical planes |
| (D) $a + b + c = 0$ and
$a^2 + b^2 + c^2 = ab + bc + ca$ | (S) the equation represent the whole of
the three dimensional space. |

- (A) (A) R (B) Q (C) P (D) S (B) (A) Q (B) R (C) P (D) S (C) (A) R (B) Q (C) S (D) P
(D) (A) R (B) P (C) Q (D) S

Ans. 5efb0e5e9fff15766e32d5fa

Sol.

$$\text{Let } \Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$\frac{-1}{2} = (a + b + c) [(a - b)^2 + (b - c)^2 + (c - a)^2]$$

- (A) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$
 $\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$
 $\Rightarrow D = 0$ and $a = b = c \neq 0$
 $\Rightarrow$ the equation represent identical planes.

- (B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
 $\Rightarrow \Delta = 0$

Since all the three planes pass through (1,1,1)

So equation of the line of intersection of these

$$\text{plane will be } \frac{x - 0}{1 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{1 - 0}$$

- (C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
 $\Rightarrow \Delta \neq 0$

$\Rightarrow$ the equations represent planes meeting
at only one point i.e. $(0,0,0)$

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow a = b = c = 0$

$\Rightarrow$ the equations represent whole of the
three dimensional space